

6. Segment Analysis

Let's do some analysis using segments like college, source, Acad plan, residency, state, sex and citizenship using field parameters.

Set up Field Parameter

1. Click "Modeling"

The screenshot shows the Power BI Desktop interface. The ribbon at the top has tabs for File, Home, Insert, Modeling, View, Optimize, and Help. The 'Modeling' tab is selected and highlighted with a purple circle. Below the ribbon, the main area displays a 'Pipeline' for the University of Faircland. The pipeline is configured for the year 2025. The main view shows two cards: 'Apps #' and 'Admits #'. The 'Apps #' card shows 254 for 2025, a decrease from 290 in the prior cycle (-12.4%). The 'Admits #' card shows 90 for 2025, an increase from 81 in the prior cycle (11.1%).

Category	2024	2025	2026
Apps #	290	254	
Admits #	81	90	

2. Click "parameter"

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Modeling

New visual calculation New measure Quick measure New column New table Calendar options Change detection **New parameter** Manage roles View as Q&A setup Language Linguistic schema

UNIVERSITY OF FABRICLAND

Pipeline

2024 2025 2026

FTC TRE

Fall Spring

APPLY Apps # 254

Prior Cycle 290 ▼ -36 (-12.4%)

Admits # 90

Prior Cycle 81 ▲ 9 (11.1%)

3. Click "Fields"

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Modeling

New visual calculation New measure Quick measure New column New table Calendar options Change detection **New parameter** Manage roles View as Q&A setup Language Linguistic schema

UNIVERSITY OF FABRICLAND

Pipeline

2024 2025 2026

FTC TRE

Fall Spring

APPLY Apps # 254

Prior Cycle 290 ▼ -36 (-12.4%)

Admits # 90

Prior Cycle 81 ▲ 9 (11.1%)

Ne 4:

Prior Cycle 69 ▼

4. Rename to "SegmentParameter"

The screenshot shows the Power BI Parameters dialog box. The 'Name' field is highlighted with a purple circle and contains the text 'SegmentParameter'. The background shows a 'Pipeline' dashboard with various filters and a 'Parameters' panel on the right.

Parameters

Add parameters to visuals and DAX expressions so people can use slicers to adjust the inputs and see different outcomes. [Learn more](#)

What will your variable adjust?

Fields

Name

SegmentParameter

Add and reorder fields

Select fields from the Fields section

Fields

- > > Measures
- > > Dim_Acad_Plan
- > > Dim_Benchmark
- > > Dim_Source
- > > Dim_State
- > > Dim_Student

5. Select "College" from the "Dim_Acad_Plan" table

The screenshot shows the Power BI Parameters dialog box. The 'Fields' list is highlighted with a purple circle, and 'Dim_Acad_Plan' is selected. The background shows a 'Pipeline' dashboard with various filters and a 'Parameters' panel on the right.

Parameters

Add parameters to visuals and DAX expressions so people can use slicers to adjust the inputs and see different outcomes. [Learn more](#)

What will your variable adjust?

Fields

Name

SegmentParameter

Add and reorder fields

Select fields from the Fields section

Fields

- > > Measures
- > > Dim_Acad_Plan
- > > Dim_Benchmark
- > > Dim_Source
- > > Dim_State
- > > Dim_Student
- > > Dim_Term
- > > Fact_Applications

☒ Add slicer to this page

Create Cancel

6. Select "Source_Desc" from the "Dim_Source" table

What will your variable adjust?

Fields

Name

SegmentParameter

Add and reorder fields

College

Fields

Search

- > Measures
- > Dim_Acad_Plan
- > Dim_Benchmark
- > Dim_Source
- > Dim_State
- > Dim_Student
- > Dim_Term
- > Fact_Applications

☒ Add slicer to this page

Create Cancel

7. Select "Acad_Plan_Code" from the "Dim_Acad_Plan" table

Parameters

What will your variable adjust?

Fields

Name

SegmentParameter

Add and reorder fields

College

Source_Desc

Fields

Search

- > Measures
- > Dim_Acad_Plan
- > Dim_Benchmark
- > Dim_Source
- > Dim_State
- > Dim_Student
- > Dim_Term
- > Fact_Applications

☒ Add slicer to this page

Create Cancel

8. Select "Residency" from the "Dim_Student" table

What will your variable adjust?

Fields

Name

SegmentParameter

Add and reorder fields

- College
- Source_Desc
- Acad_Plan_Code

Fields

- > Measures
- > Dim_Acad_Plan
- > Dim_Benchmark
- > Dim_Source
- > Dim_State
- > **Dim_Student**
- > Dim_Term
- > Fact_Applications

☒ Add slicer to this page

Create Cancel

9. Select "State_Name" from the "Dim_State" table

What will your variable adjust?

Fields

Name

SegmentParameter

Add and reorder fields

- College
- Source_Desc
- Acad_Plan_Code
- Residency

Fields

- > Measures
- > Dim_Acad_Plan
- > Dim_Benchmark
- > Dim_Source
- > **Dim_State**
- > Dim_Student
- > Dim_Term
- > Fact_Applications

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Create Cancel

10. Select "Sex" and "Citizenship" from the "Dim_Student" table

What will your variable adjust?

Fields

Name

SegmentParameter

Add and reorder fields

- College
- Source_Desc
- Acad_Plan_Code
- Residency
- State_Name

Fields

- > Measures
- > Dim_Acad_Plan
- > Dim_Benchmark
- > Dim_Source
- > Dim_State
- > **Dim_Student**
- > Dim_Term
- > Fact_Applications

☒ Add slicer to this page

Create Cancel

11. Click "Create" to finalize the parameter creation

What will your variable adjust?

Fields

Name

SegmentParameter

Add and reorder fields

- College
- Source_Desc
- Acad_Plan_Code
- Residency
- State_Name
- Sex
- Citizenship

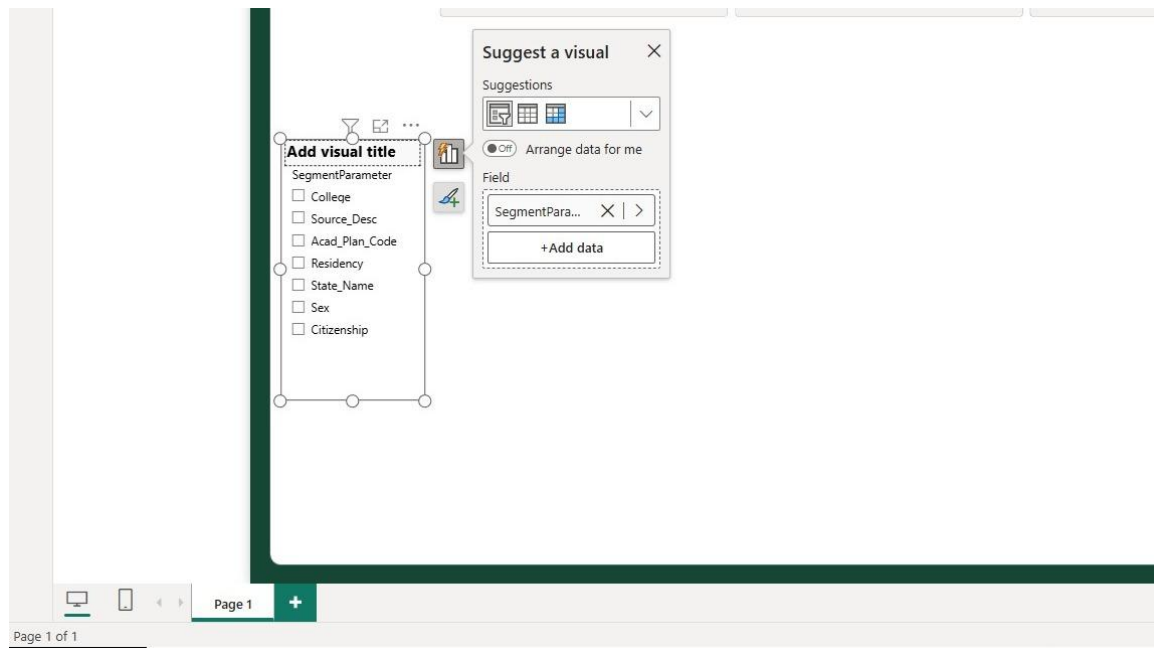
Fields

- > Dim_Student
 - ☒ Citizenship
 - ☐ First_Name
 - ☐ Full_Name
 - ☐ HiF_Status
 - ☐ Last_Name
 - ☒ Residency
 - ☒ Sex
 - ☐ State_Code
 - ☐ StudentID
- > Dim_Term
- > Fact_Applications

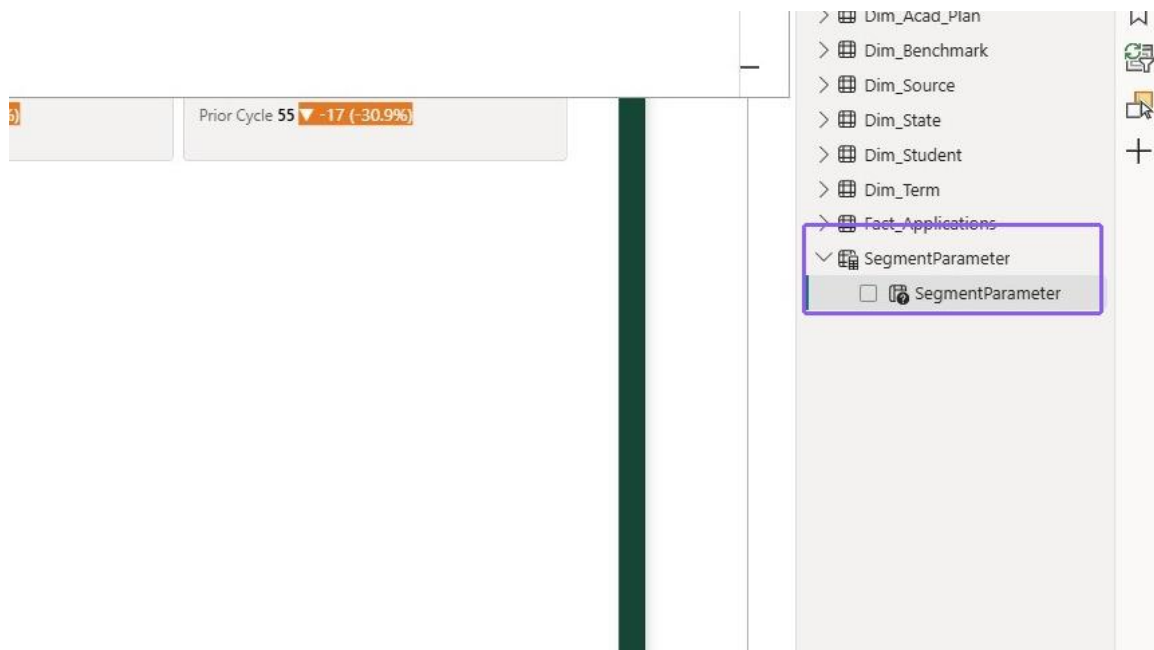
☒ Add slicer to this page

Create Cancel

12. Verify the new parameter and slicer on the report canvas



13. You can also verify by checking the new table added "SegmentParameter"



Rename Parameters

14. Click "SegmentParameter"

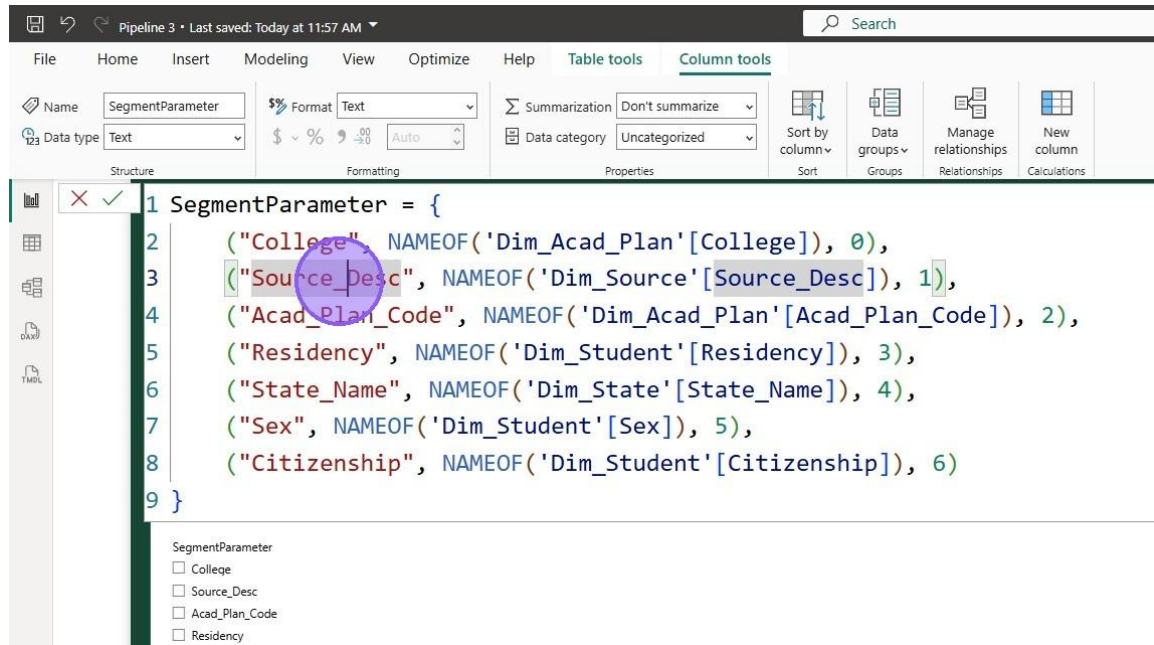
The screenshot shows the Power BI interface. On the left, there are two cards: 'Net Deposits #' with a value of 43 and 'Enrolled #' with a value of 38. On the right, the 'Filters' pane is open, showing a list of parameters. The 'SegmentParameter' parameter is highlighted with a red circle.

15. Edit in the Formula bar

The screenshot shows the Power BI interface with the 'SegmentParameter' parameter selected in the 'Column tools' pane. The 'Formula bar' is open, showing the DAX formula for 'SegmentParameter'. The formula is a table with columns 'College', 'Source_Desc', 'Acad_Plan_Code', 'Residency', 'State_Name', 'Sex', and 'Citizenship'. The 'College' column is highlighted with a red circle.

```
1 SegmentParameter = {
2     ("College", NAMEOF('Dim_Acad_Plan'[College]), 0),
3     ("Source_Desc", NAMEOF('Dim_Source'[Source_Desc]), 1),
4     ("Acad_Plan_Code", NAMEOF('Dim_Acad_Plan'[Acad_Plan_Code]), 2),
5     ("Residency", NAMEOF('Dim_Student'[Residency]), 3),
6     ("State_Name", NAMEOF('Dim_State'[State_Name]), 4),
7     ("Sex", NAMEOF('Dim_Student'[Sex]), 5),
8     ("Citizenship", NAMEOF('Dim_Student'[Citizenship]), 6)
9 }
```

16. Source



Pipeline 3 • Last saved: Today at 11:57 AM

File Home Insert Modeling View Optimize Help Table tools Column tools

Name SegmentParameter Format Text Summarization Don't summarize Data category Uncategorized Sort by column Sort Data groups Groups Manage relationships Relationships New column Calculations

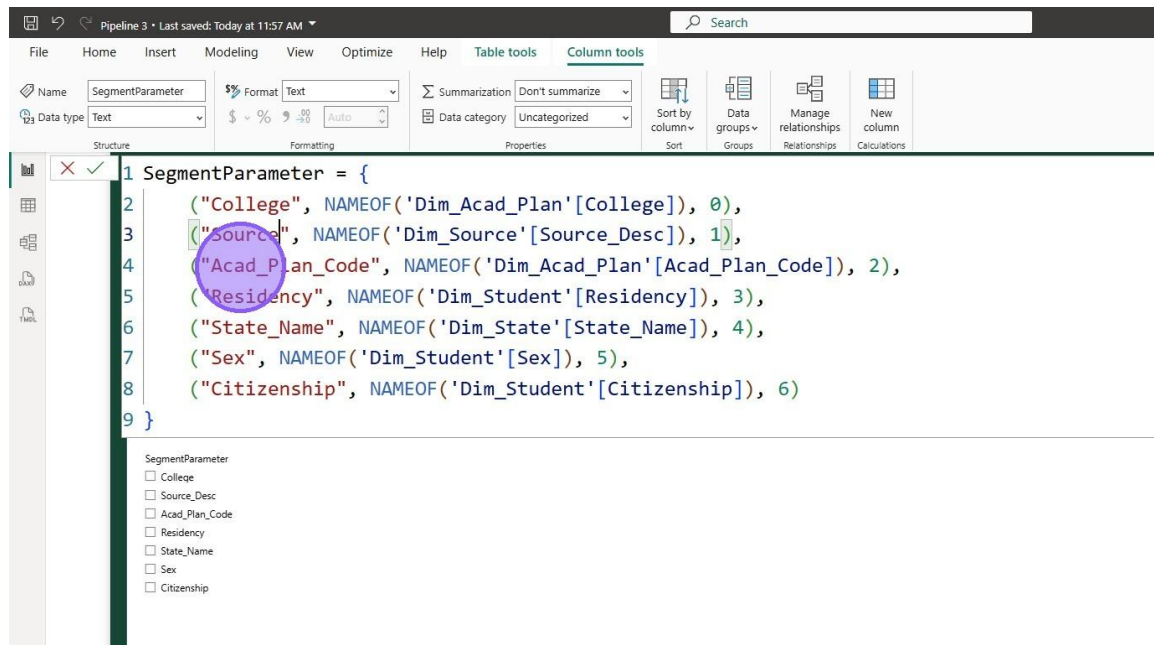
Structure Formatting Properties

```
1 SegmentParameter = {
2   ("College", NAMEOF('Dim_Acad_Plan'[College]), 0),
3   ("Source_Desc", NAMEOF('Dim_Source'[Source_Desc]), 1),
4   ("Acad_Plan_Code", NAMEOF('Dim_Acad_Plan'[Acad_Plan_Code]), 2),
5   ("Residency", NAMEOF('Dim_Student'[Residency]), 3),
6   ("State_Name", NAMEOF('Dim_State'[State_Name]), 4),
7   ("Sex", NAMEOF('Dim_Student'[Sex]), 5),
8   ("Citizenship", NAMEOF('Dim_Student'[Citizenship]), 6)
9 }
```

SegmentParameter

- ☐ College
- ☐ Source_Desc
- ☐ Acad_Plan_Code
- ☐ Residency

17. Acad Plan



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File Home Insert Modeling View Optimize Help Table tools Column tools

Name SegmentParameter Format Text Summarization Don't summarize Data category Uncategorized Sort by column Sort Data groups Groups Manage relationships Relationships New column Calculations

Structure Formatting Properties

```
1 SegmentParameter = {
2   ("College", NAMEOF('Dim_Acad_Plan'[College]), 0),
3   ("Source", NAMEOF('Dim_Source'[Source_Desc]), 1),
4   ("Acad_Plan_Code", NAMEOF('Dim_Acad_Plan'[Acad_Plan_Code]), 2),
5   ("Residency", NAMEOF('Dim_Student'[Residency]), 3),
6   ("State_Name", NAMEOF('Dim_State'[State_Name]), 4),
7   ("Sex", NAMEOF('Dim_Student'[Sex]), 5),
8   ("Citizenship", NAMEOF('Dim_Student'[Citizenship]), 6)
9 }
```

SegmentParameter

- ☐ College
- ☐ Source_Desc
- ☐ Acad_Plan_Code
- ☐ Residency
- ☐ State_Name
- ☐ Sex
- ☐ Citizenship

18. State

The screenshot shows a data modeling tool interface. The top menu bar includes File, Home, Insert, Modeling, View, Optimize, Help, Table tools, and Column tools. The 'Column tools' tab is active, showing options for Name, Data type, Format, Summarization, Data category, Sort by column, Data groups, Manage relationships, and New column. The 'Name' field is set to 'SegmentParameter' and the 'Data type' is 'Text'. The 'Format' is set to 'Text' and the 'Summarization' is 'Don't summarize'. The 'Data category' is 'Uncategorized'. The 'Sort by column' is 'Sort', 'Data groups' is 'Groups', 'Manage relationships' is 'Relationships', and 'New column' is 'Calculations'. The main area shows a column definition for 'SegmentParameter' with the following code:

```
1 SegmentParameter = {  
2     ("College", NAMEOF('Dim_Acad_Plan'[College]), 0),  
3     ("Source", NAMEOF('Dim_Source'[Source_Desc]), 1),  
4     ("Acad Plan", NAMEOF('Dim_Acad_Plan'[Acad_Plan_Code]), 2),  
5     ("Residency", NAMEOF('Dim_Student'[Residency]), 3),  
6     ("State Name", NAMEOF('Dim_State'[State_Name]), 4),  
7     ("Sex", NAMEOF('Dim_Student'[Sex]), 5),  
8     ("Citizenship", NAMEOF('Dim_Student'[Citizenship]), 6)  
9 }
```

The 'State Name' entry is highlighted with a purple circle. Below the code, a list of segment parameters is shown:

- ☐ College
- ☐ Source_Desc
- ☐ Acad_Plan_Code
- ☐ Residency
- ☐ State_Name
- ☐ Sex
- ☐ Citizenship

19. These are the final names

The screenshot shows the same data modeling tool interface as in the previous image. The 'Column tools' tab is active. The main area shows the final column definition for 'SegmentParameter' with the following code:

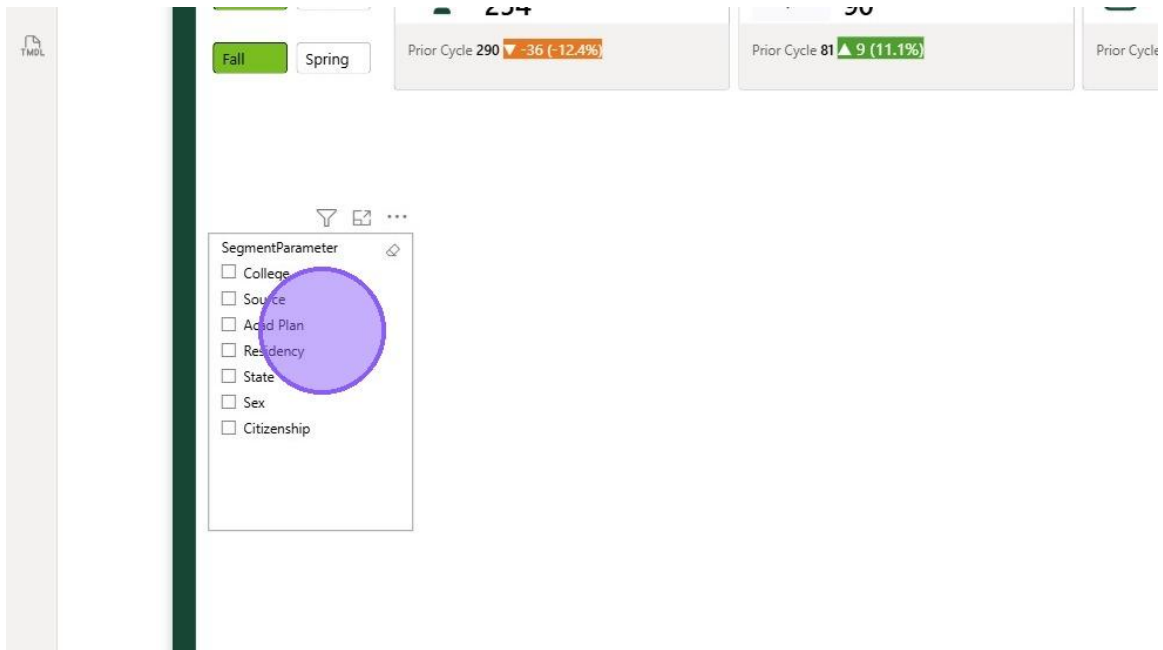
```
1 SegmentParameter = {  
2     ("College", NAMEOF('Dim_Acad_Plan'[College]), 0),  
3     ("Source", NAMEOF('Dim_Source'[Source_Desc]), 1),  
4     ("Acad Plan", NAMEOF('Dim_Acad_Plan'[Acad_Plan_Code]), 2),  
5     ("Residency", NAMEOF('Dim_Student'[Residency]), 3),  
6     ("State", NAMEOF('Dim_State'[State_Name]), 4),  
7     ("Sex", NAMEOF('Dim_Student'[Sex]), 5),  
8     ("Citizenship", NAMEOF('Dim_Student'[Citizenship]), 6)  
9 }
```

The 'State' entry is highlighted with a purple box. Below the code, a list of segment parameters is shown:

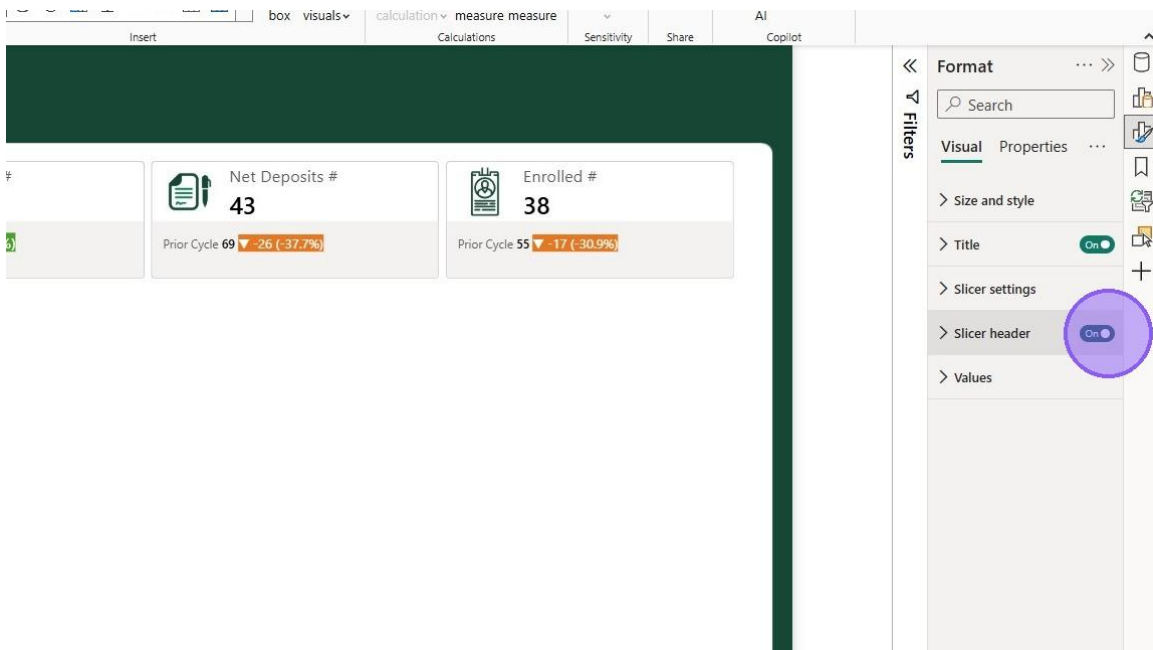
- ☐ College
- ☐ Source_Desc
- ☐ Acad_Plan_Code
- ☐ Residency
- ☐ State_Name
- ☐ Sex
- ☐ Citizenship

Slicer Formatting

20. Click "SegmentParameter"



21. Turn off the Slicer Header



22. Turn off the title

The screenshot shows the Power BI Desktop interface. The main view displays two cards: "Net Deposits #" with a value of 43 and "Enrolled #" with a value of 38. Both cards show a change from the prior cycle. The "Format" pane on the right is open, and the "Title" property is highlighted with a purple circle, indicating it is turned off.

Card	Value	Prior Cycle	Change
Net Deposits #	43	69	-26 (-37.7%)
Enrolled #	38	55	-17 (-30.9%)

23. Click "Button slicer" to change the slicer type

The screenshot shows the Power BI Desktop interface. The main view displays the same two cards as in the previous screenshot. The "Build" pane on the right is open, and the "Button slicer" option is highlighted with a purple circle, indicating it is selected.

Card	Value	Prior Cycle	Change
Net Deposits #	43	69	-26 (-37.7%)
Enrolled #	38	55	-17 (-30.9%)

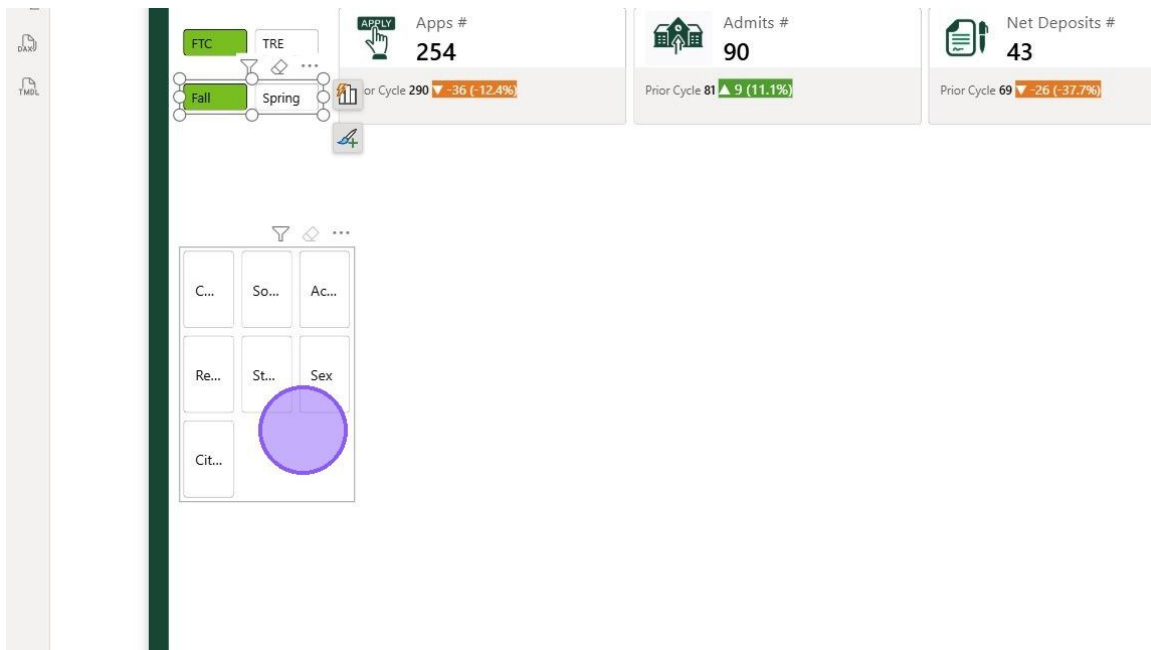
24. Click the already formatted term slicer

The screenshot shows the Power BI interface with the 'Pipeline' report. The top ribbon includes the 'Format painter' button, which is highlighted with a purple circle. The report displays data for the 'University of Fabricland' for the years 2024, 2025, and 2026. It includes a 'Term' slicer with 'Fall' and 'Spring' options, and two KPI cards: 'Apps #' (254) and 'Admits #' (90). The 'Apps #' card shows a change from the prior cycle of 290 to 254, a decrease of 36 (-12.4%). The 'Admits #' card shows a change from the prior cycle of 81 to 90, an increase of 9 (11.1%).

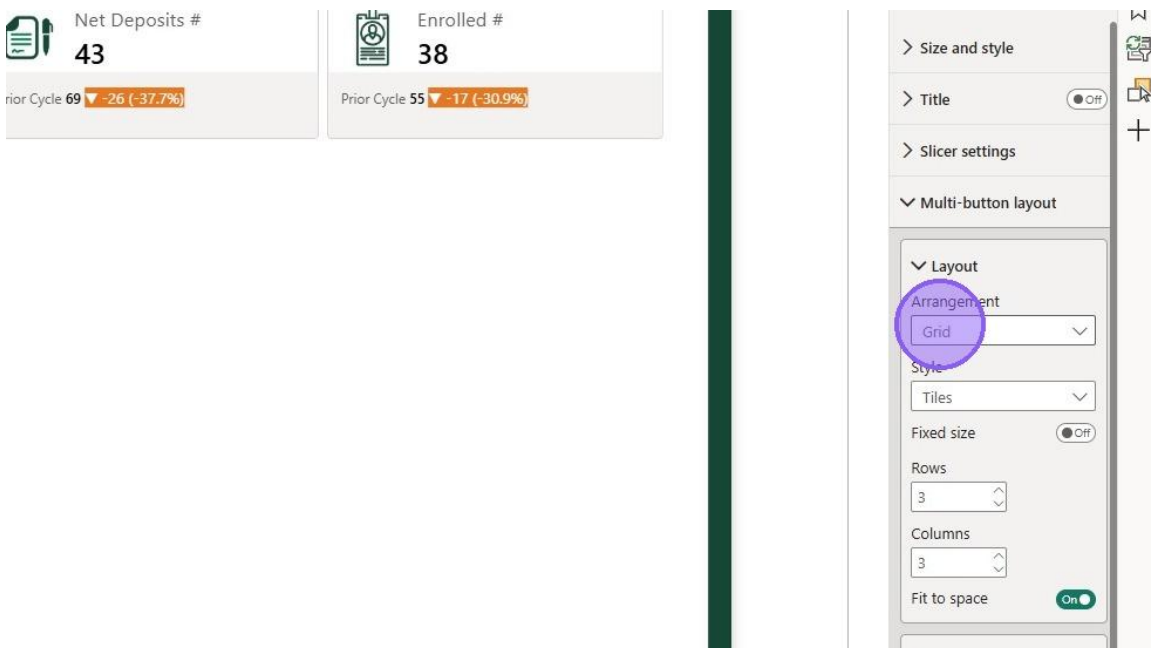
25. Click "Format painter"

The screenshot shows the Power BI interface with the 'Pipeline' report. The top ribbon includes the 'Format painter' button, which is highlighted with a purple circle. The report displays data for the 'University of Fabricland' for the years 2024, 2025, and 2026. It includes a 'Term' slicer with 'Fall' and 'Spring' options, and two KPI cards: 'Apps #' (254) and 'Admits #' (90). The 'Apps #' card shows a change from the prior cycle of 290 to 254, a decrease of 36 (-12.4%). The 'Admits #' card shows a change from the prior cycle of 81 to 90, an increase of 9 (11.1%).

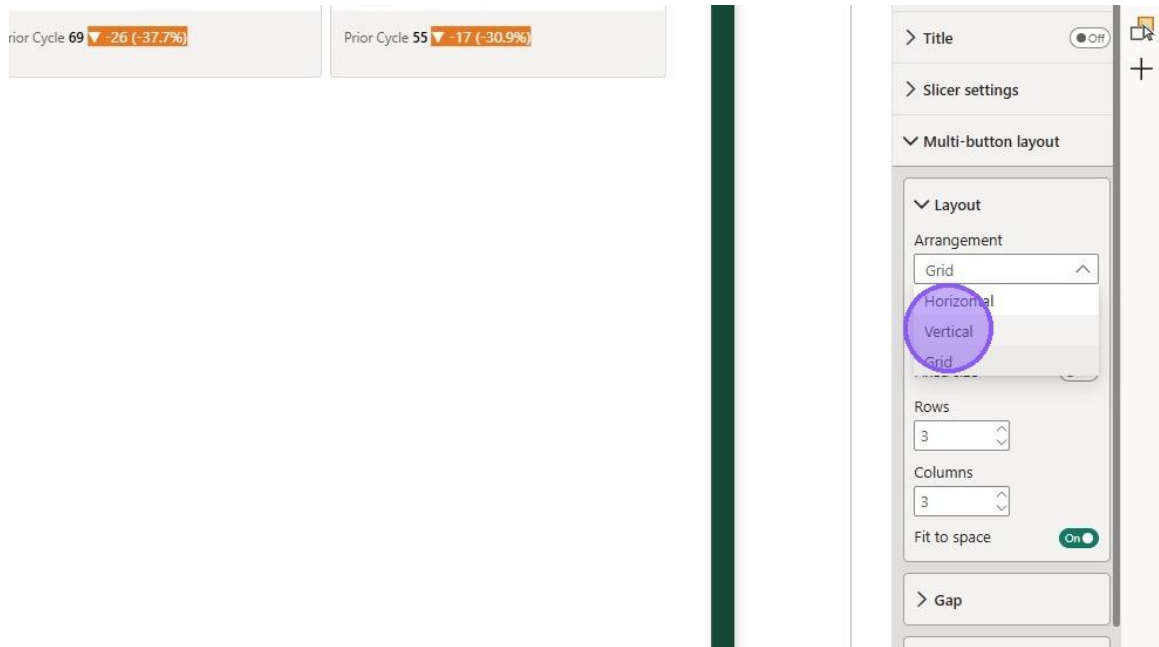
26. Then click on the segment slicer



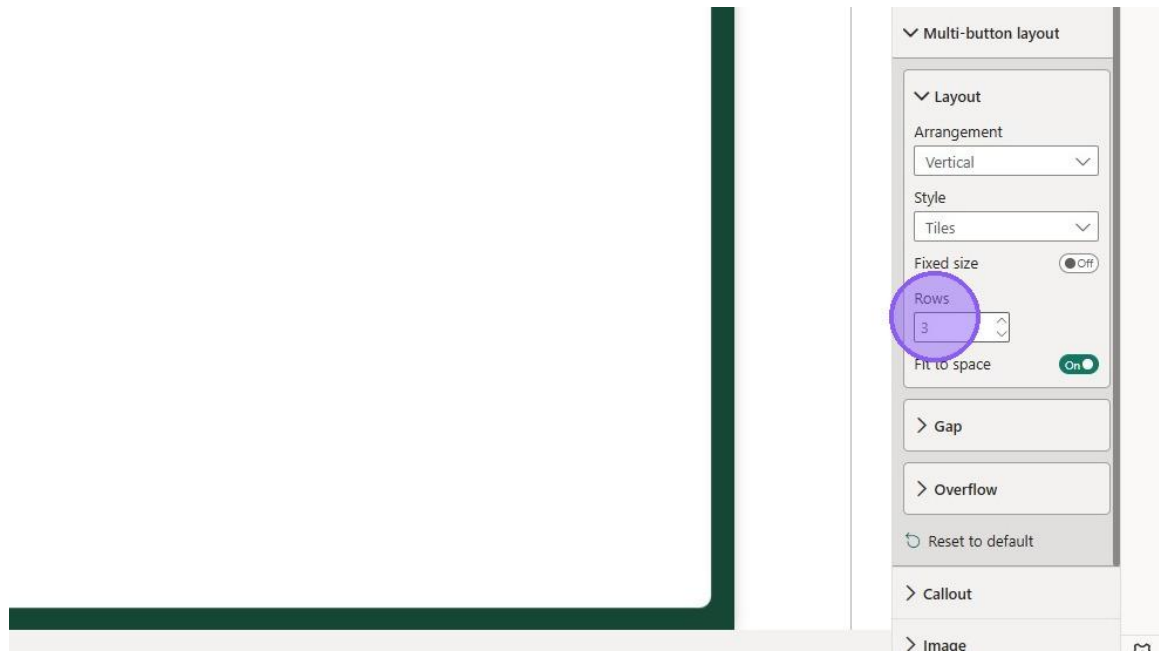
27. Under Multi Button Layout, Click "Arrangement"



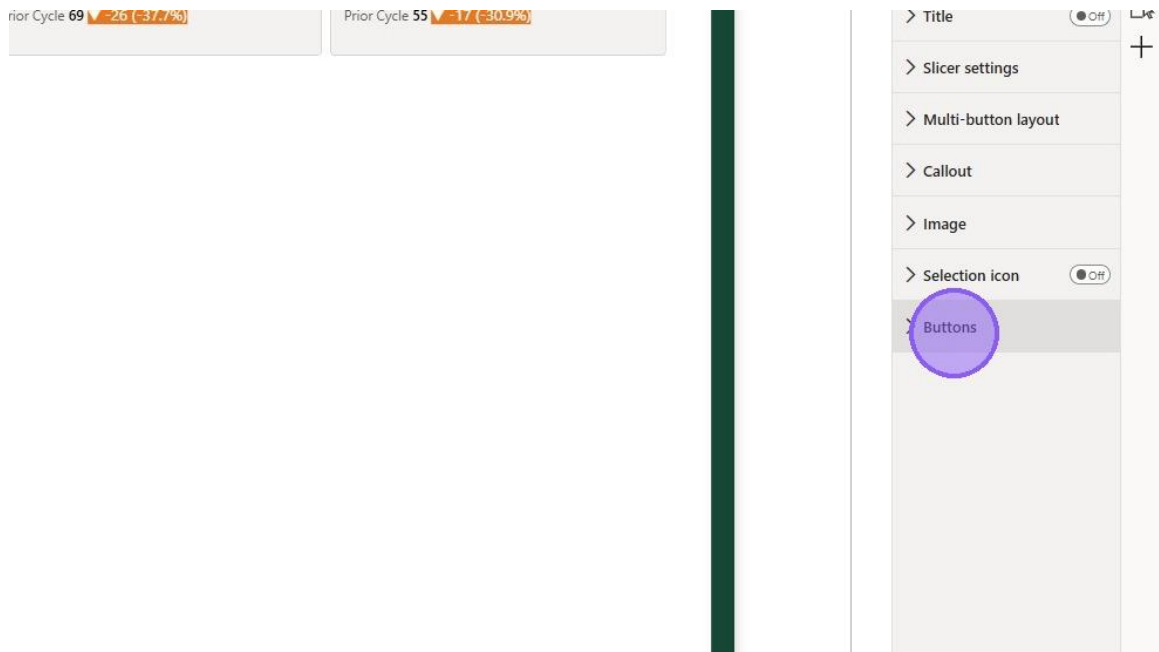
28. Select Vertical



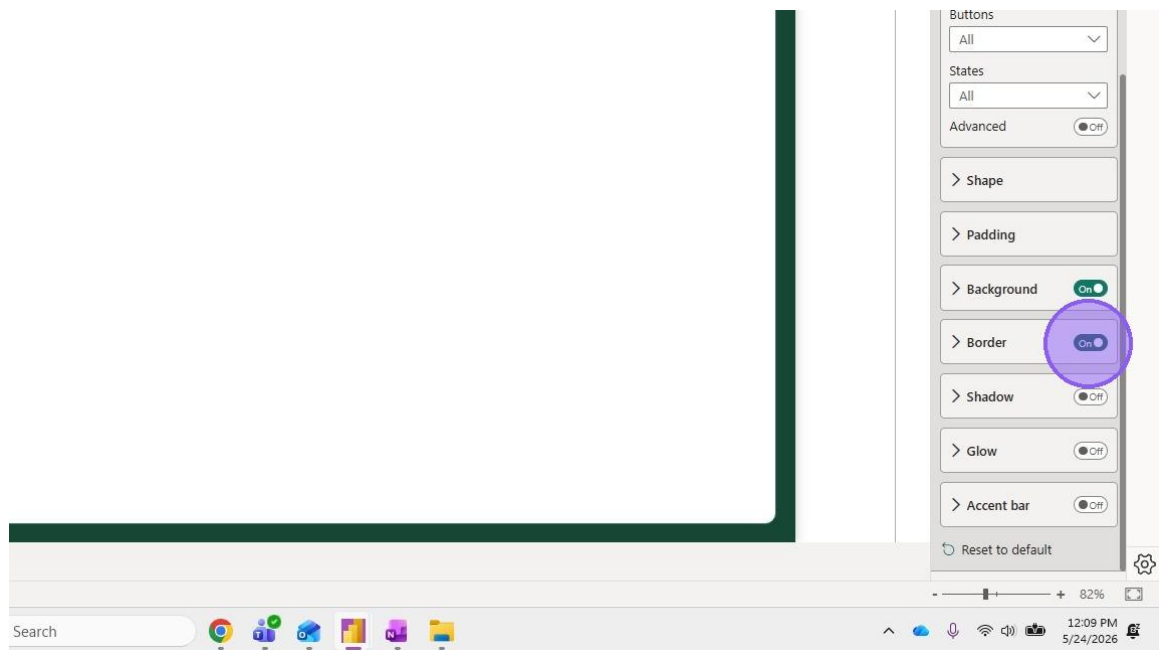
29. Type 7 under Rows



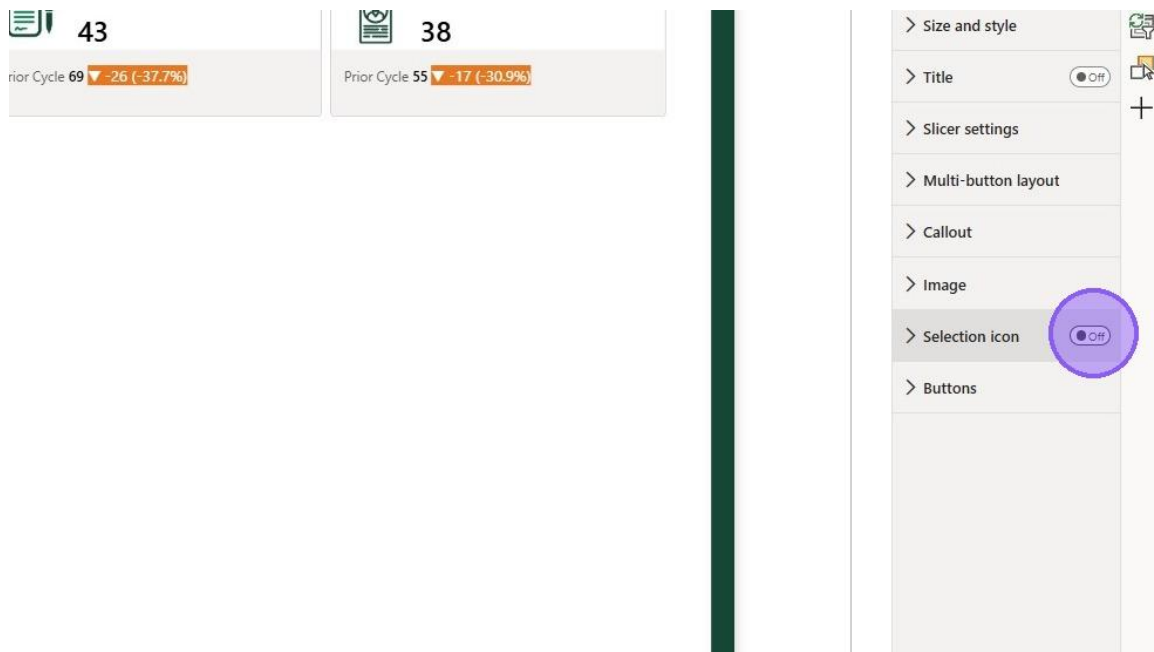
30. Click "Buttons"



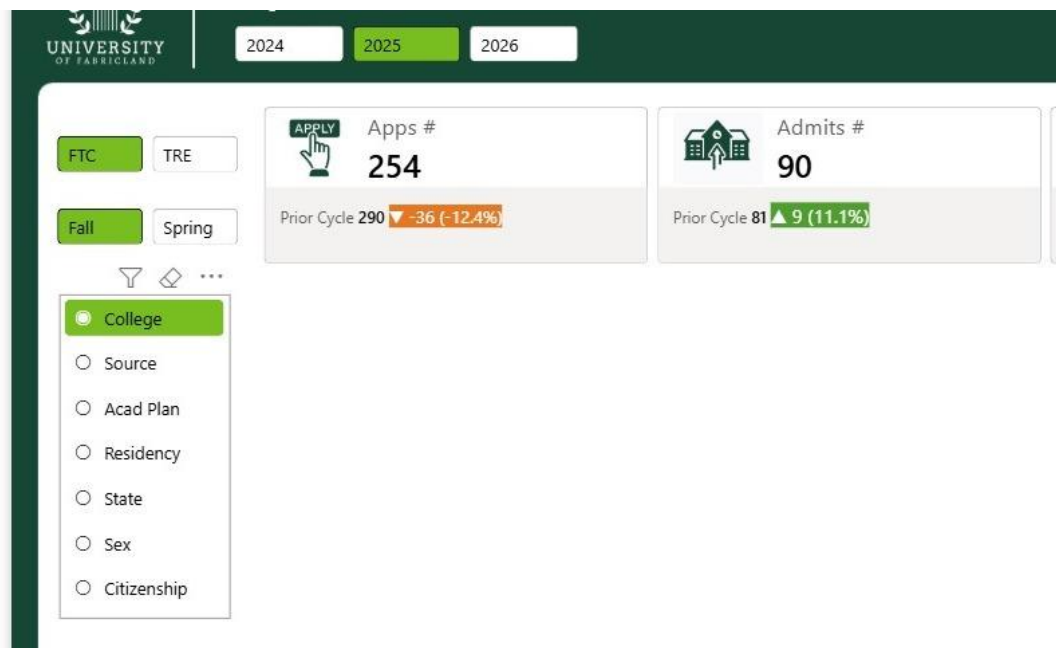
31. Switch off the border



32. Switch on the Selection icon

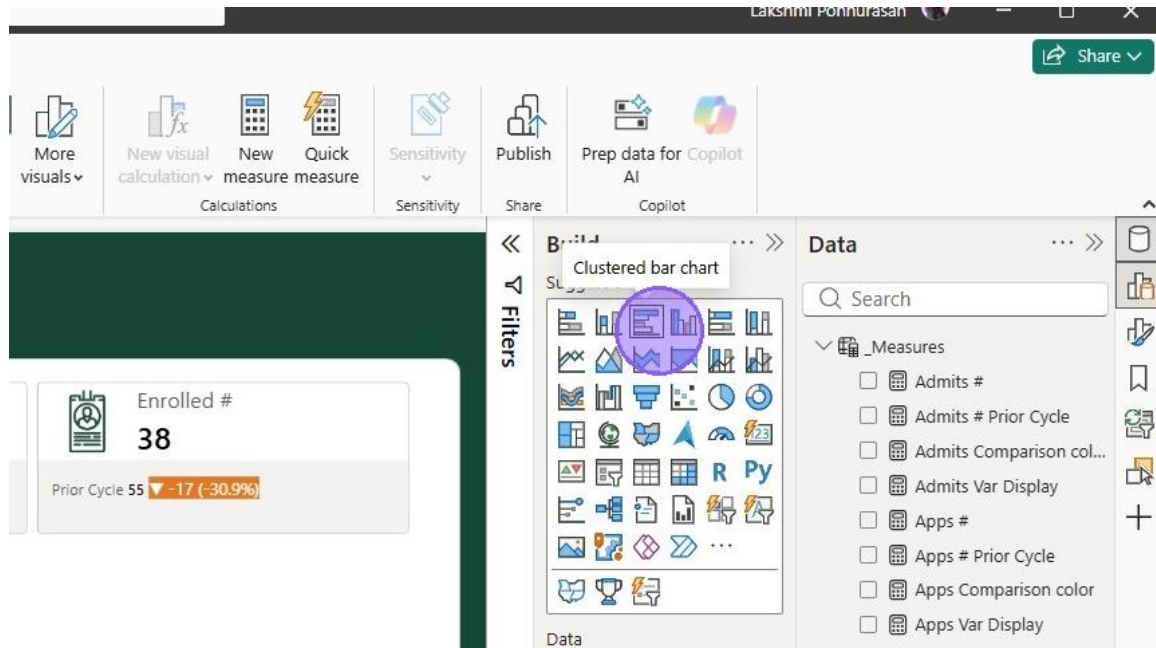


33. This is how the segment slicer should look like.



Bar Chart

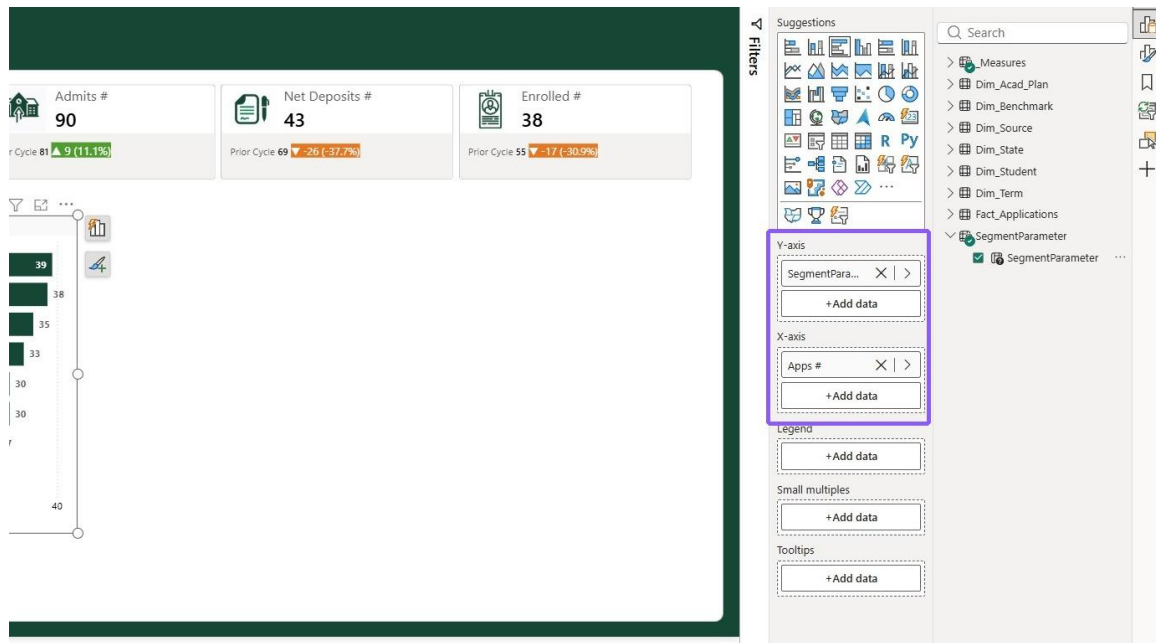
34. Click "Clustered bar chart"



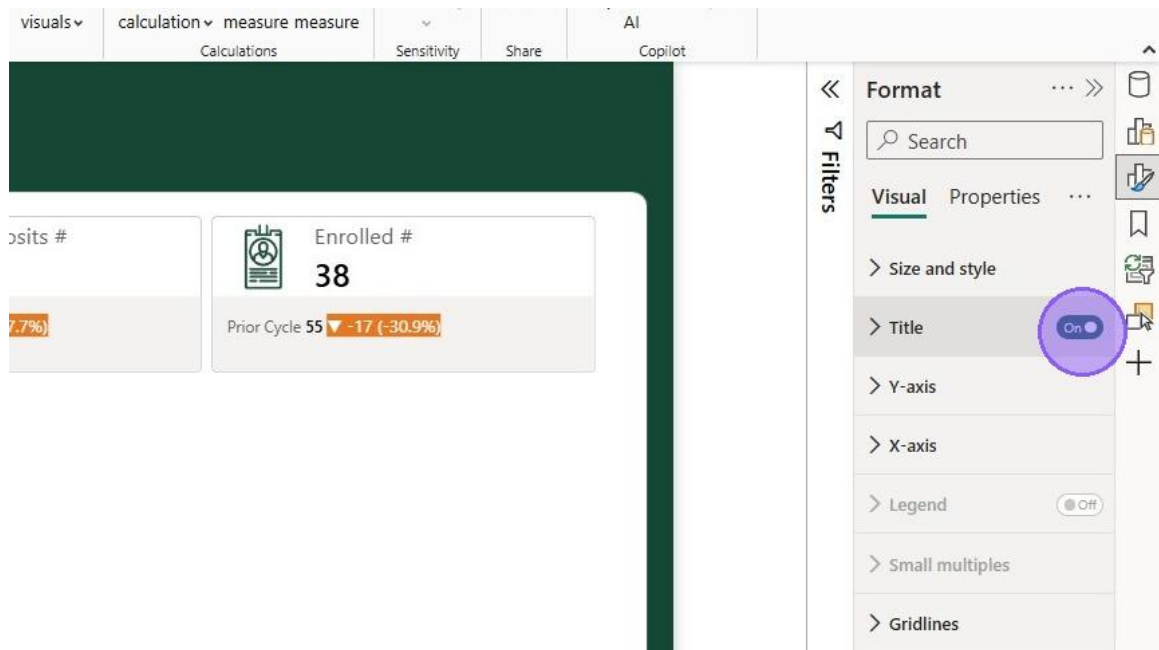
35. Drag and drop

Apps # into X Axis

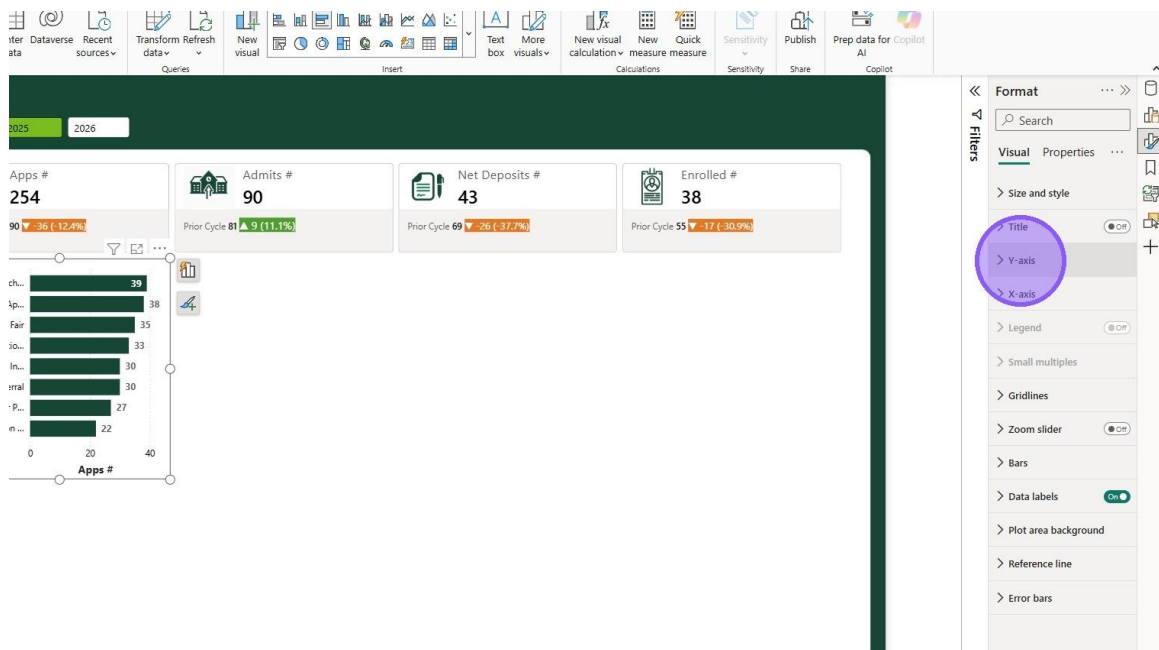
SegmentParameter into Y Axis



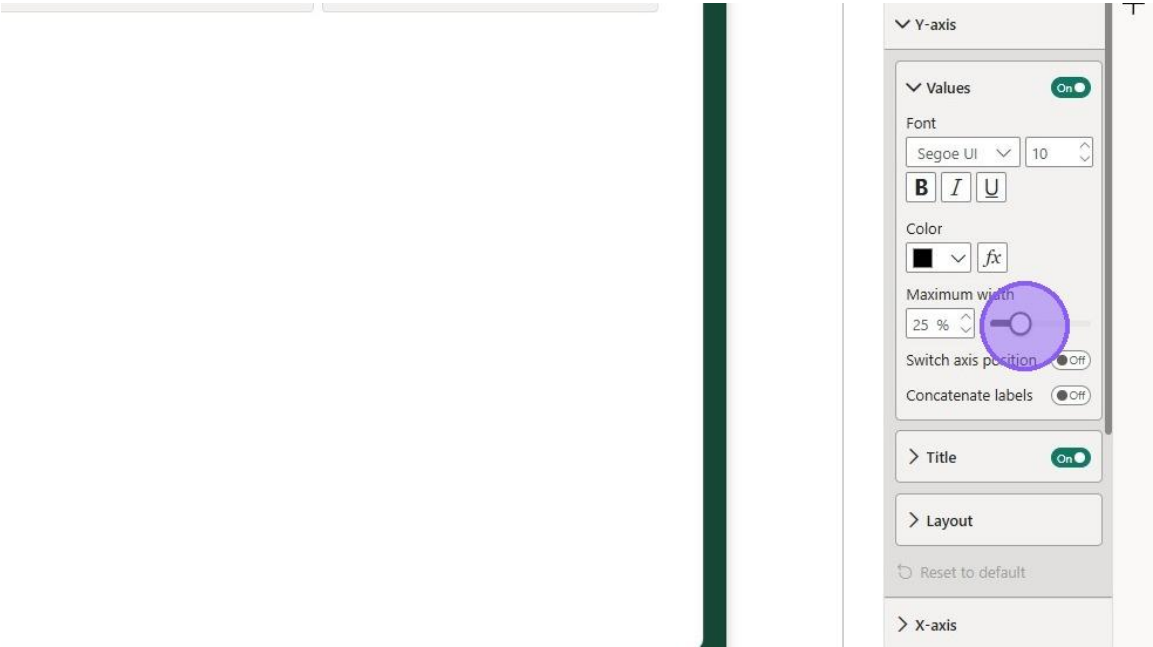
36. Switch off Title



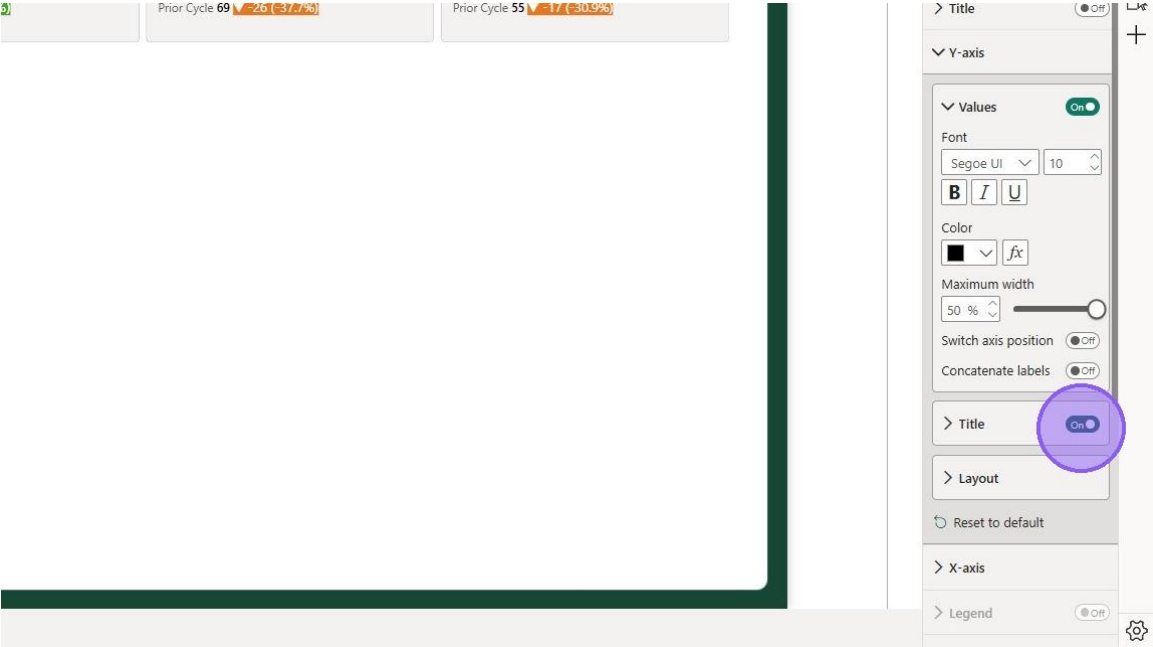
37. Click "Y-axis"



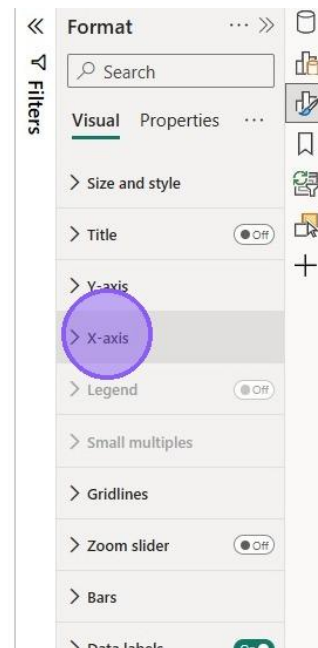
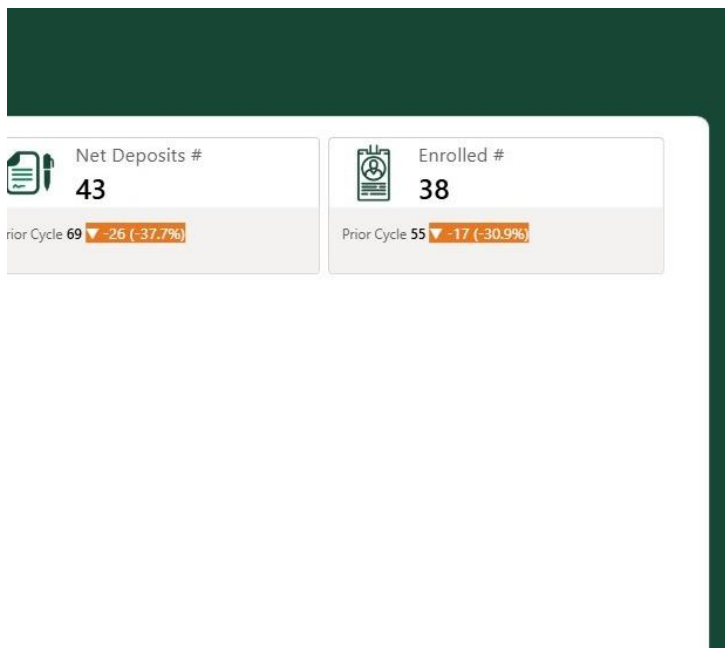
38. Increase Maximum Width to 50%



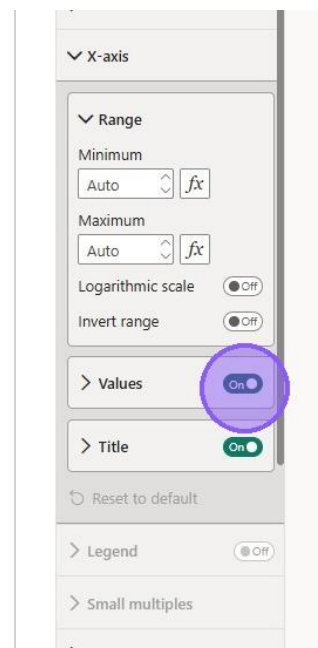
39. Switch off title



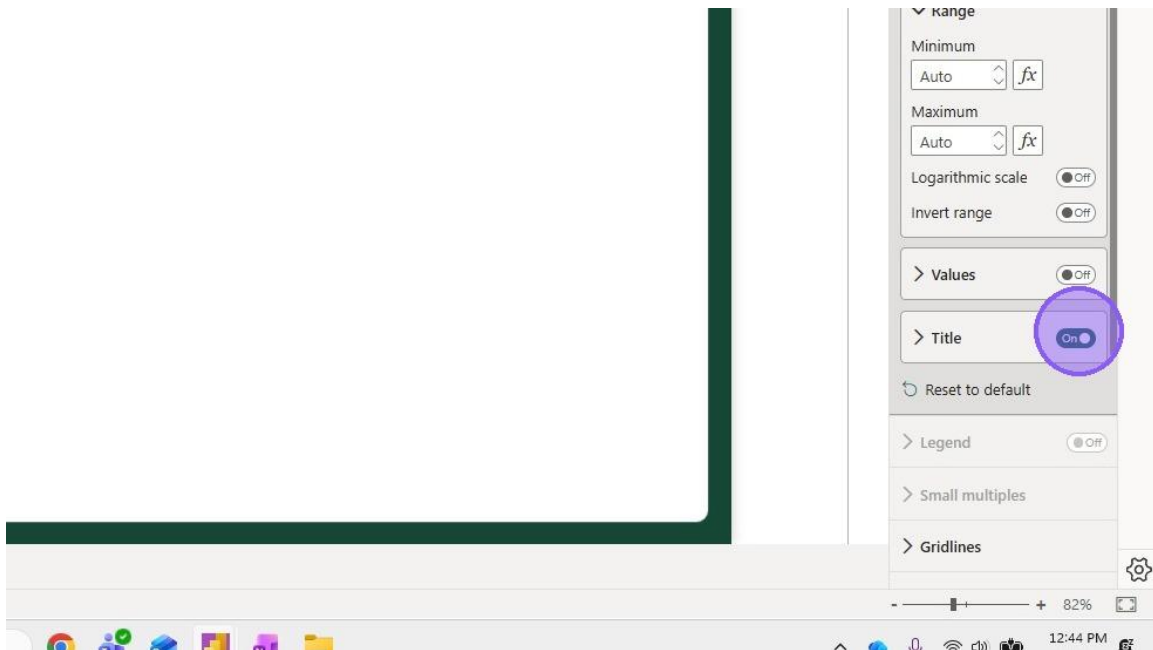
40. Click "X-axis"



41. Switch off values



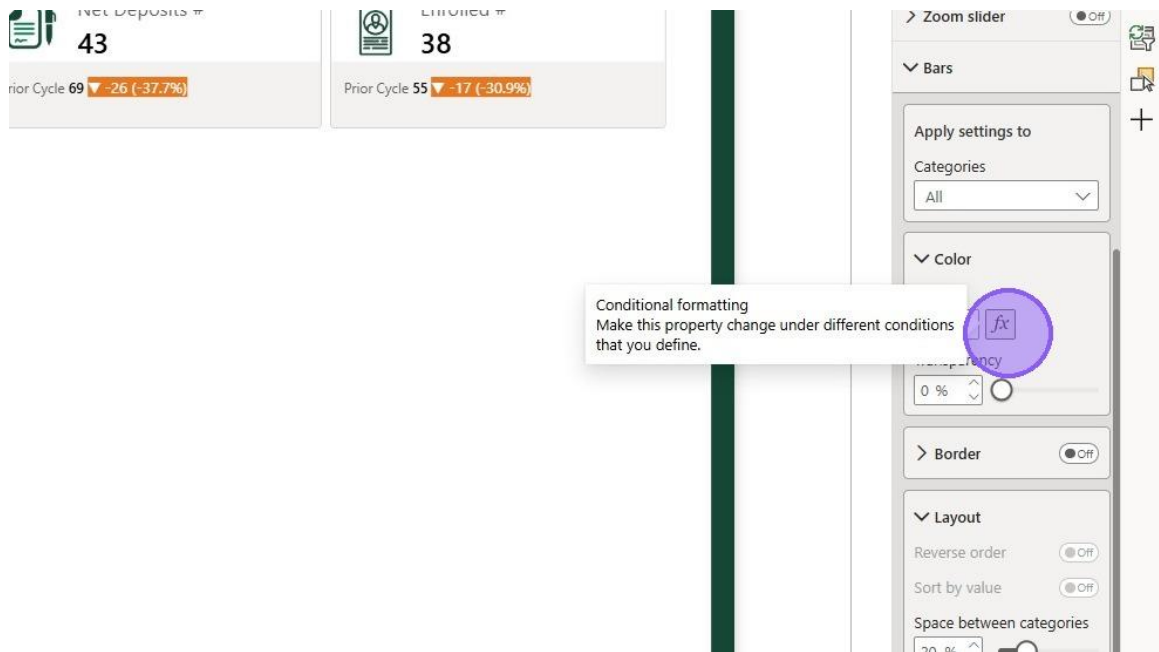
42. Switch off title



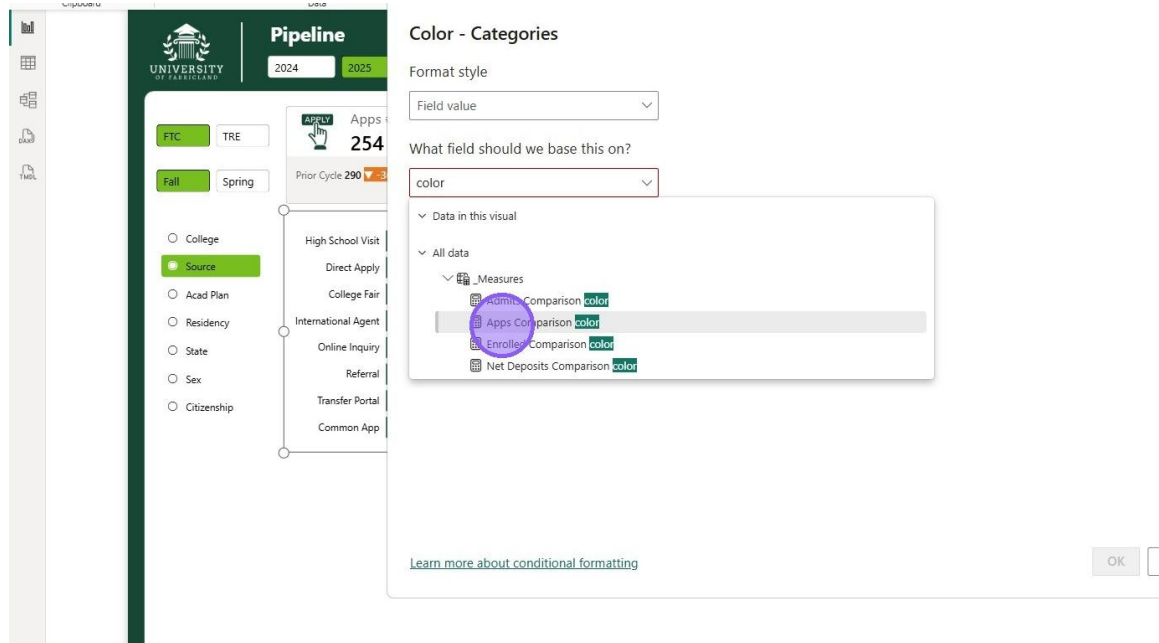
43. Click "Bars"



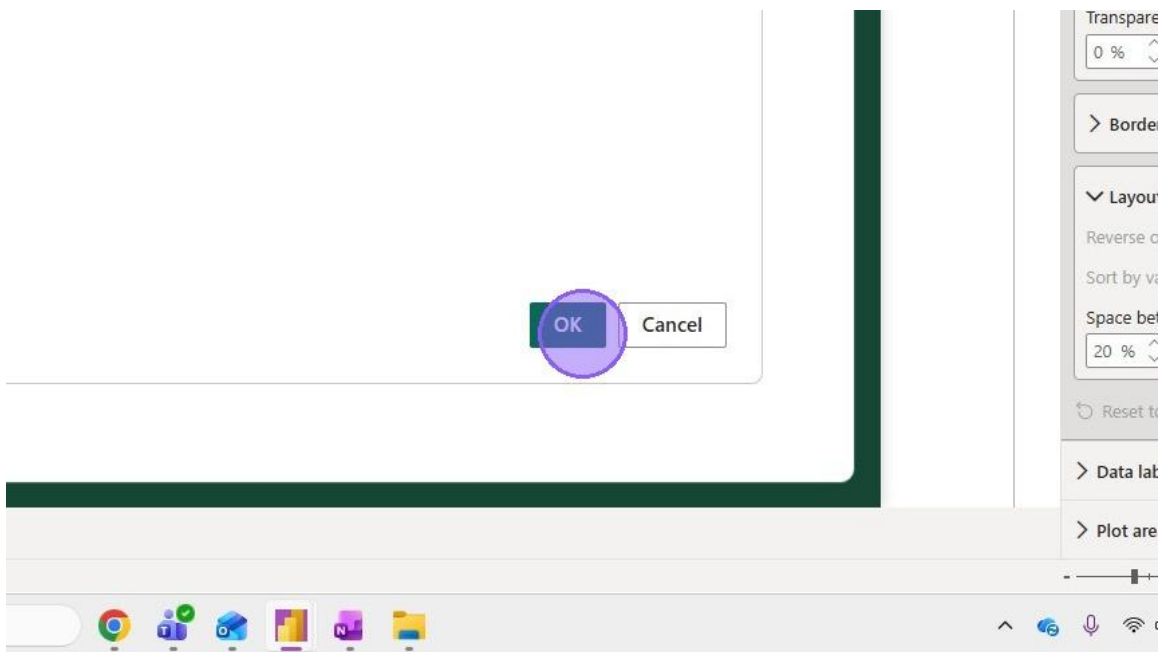
44. Select fx next to color



45. Select Field Value format style and then choose the Apps Comparison Color measure from the drop down



46. Click "OK"



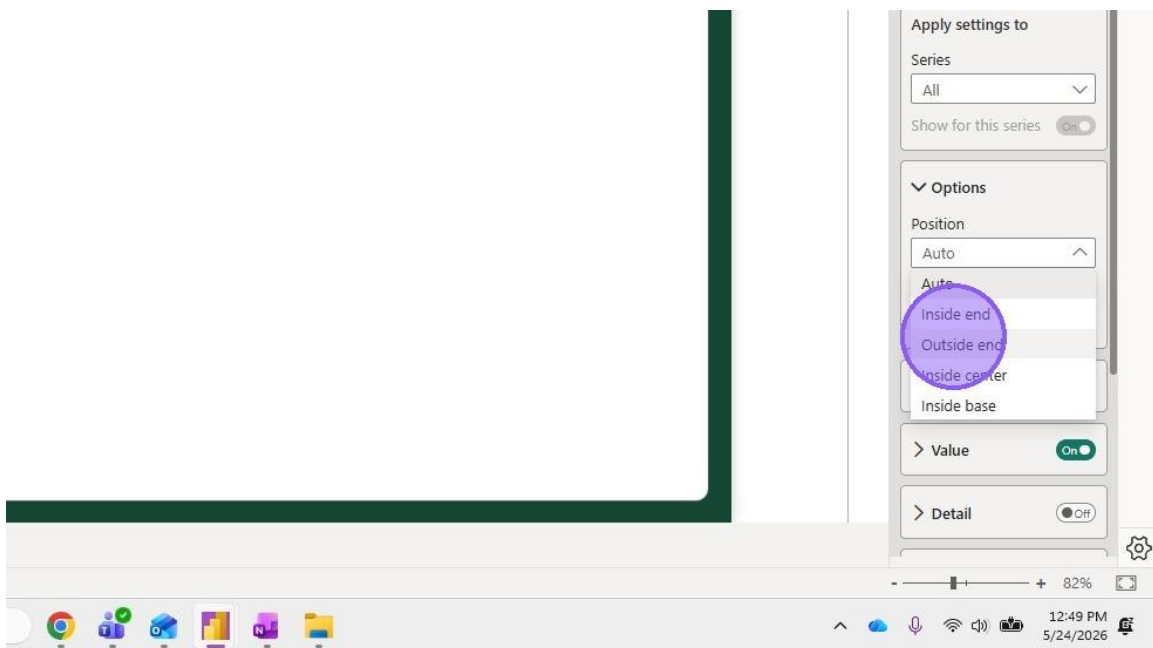
47. Click "Space between categories" and increase to 70%



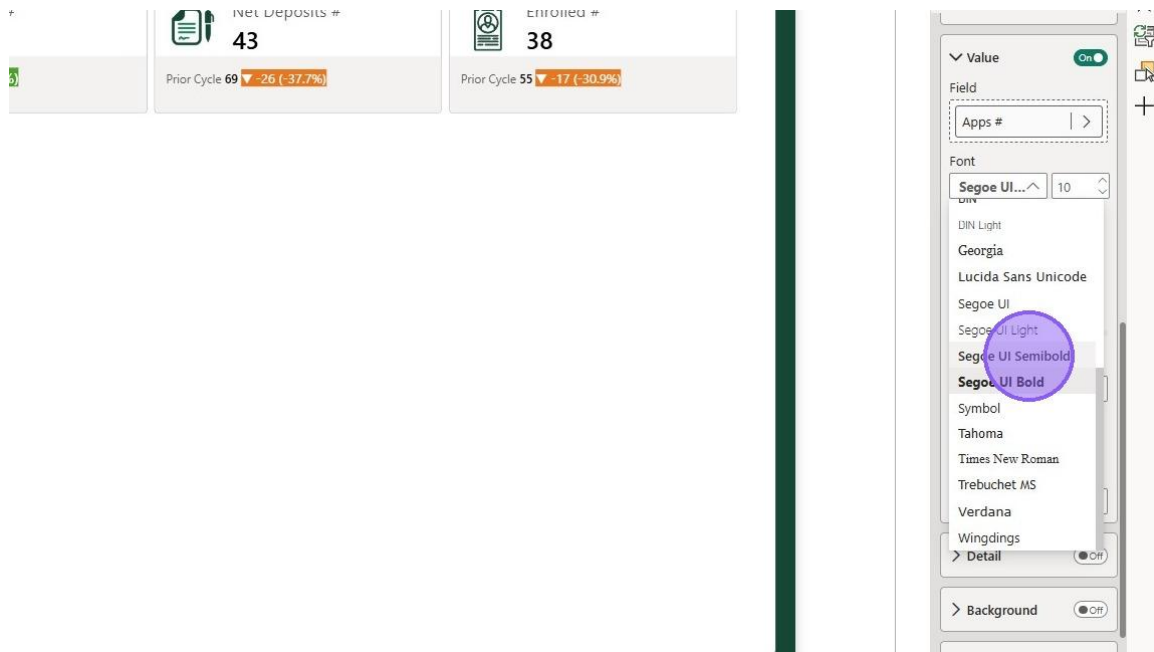
48. Click "Data labels"



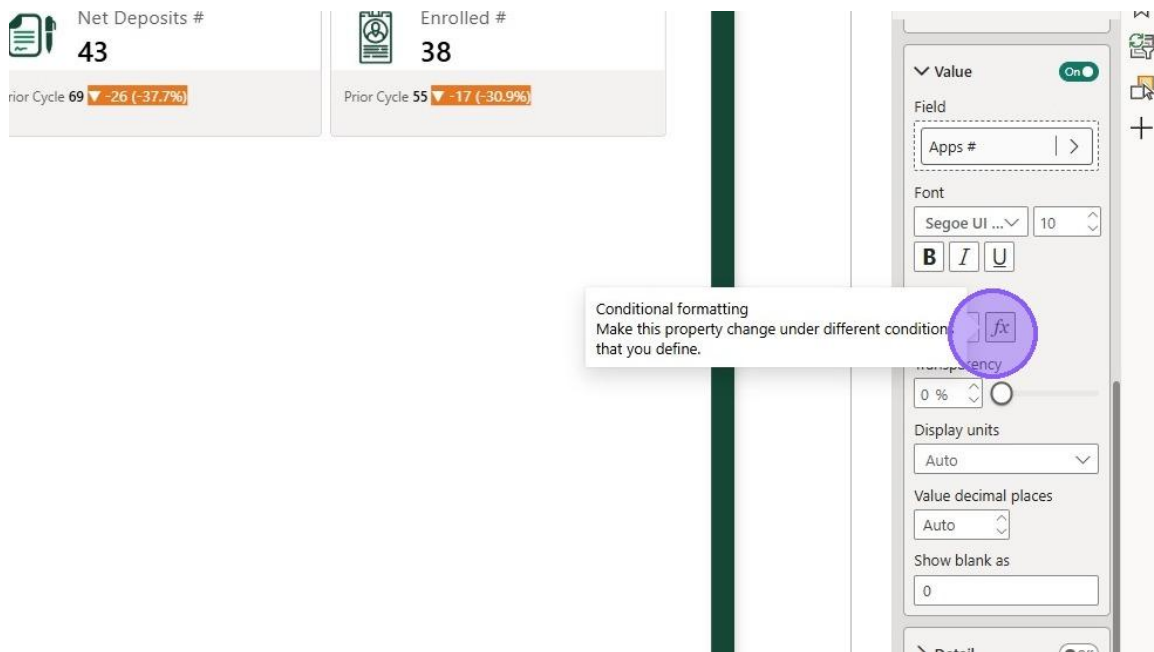
49. Click "Outside end" under Position



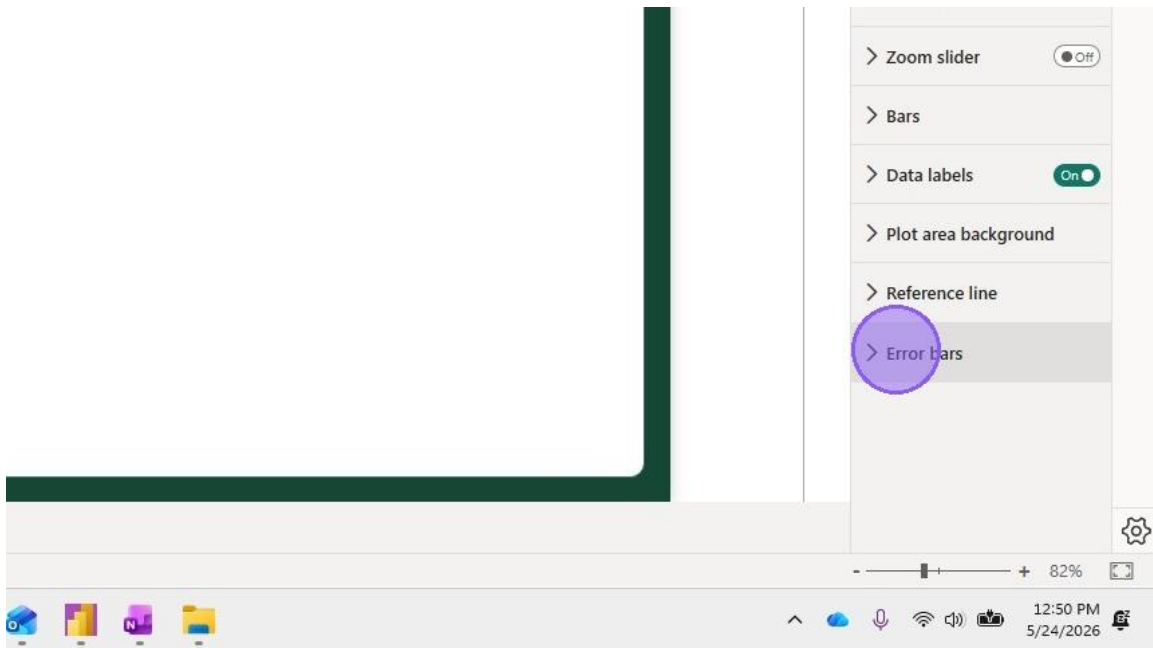
50. Click "Segoe UI Semibold" under Font



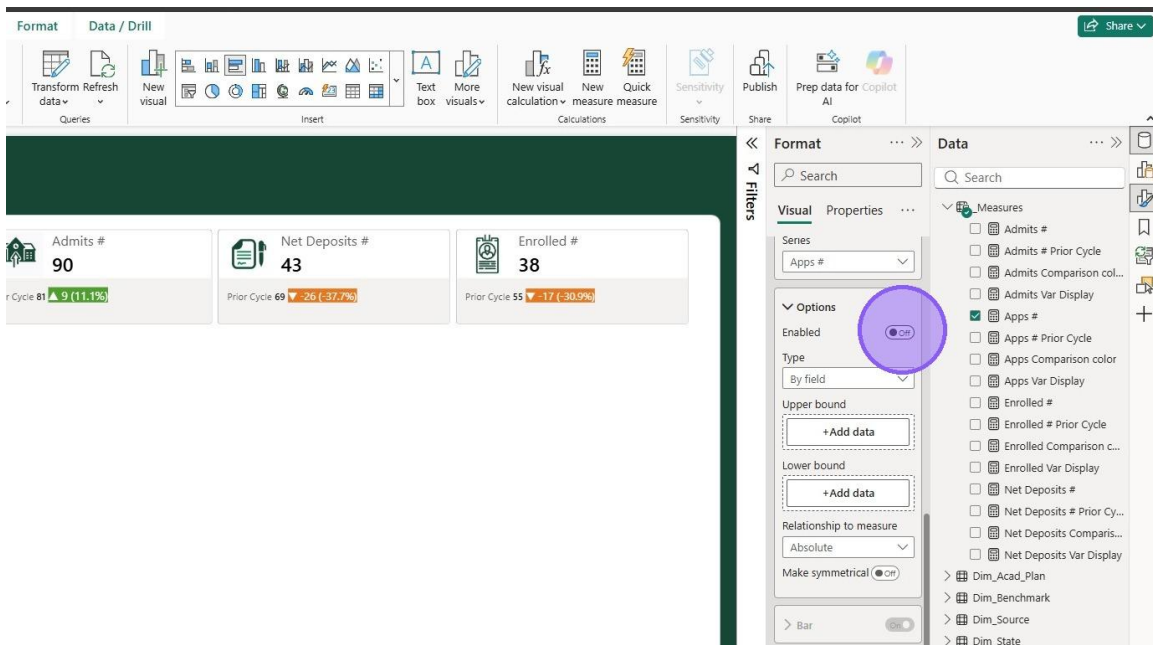
51. Select fx next to color



52. Select Field Value format style and then choose the Apps Comparison Color measure from the drop down like we did before. Click Ok
53. Click "Error bars"



54. Enable the toggle button



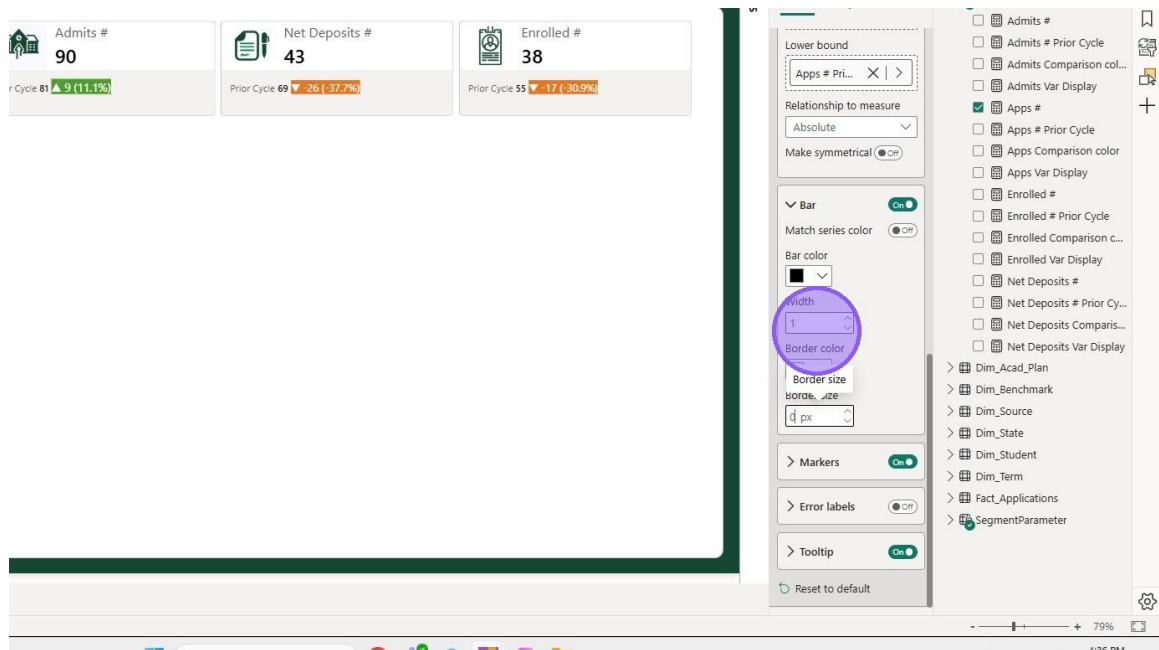
55. Drag and drop "Apps # Prior Cycle" into the Upper and Lower Bound buckets

The screenshot shows the Power BI Desktop interface. The main view displays three cards: 'Admits #' with a value of 90, 'Net Deposits #' with a value of 43, and 'Enrolled #' with a value of 38. Each card shows a change from the 'Prior Cycle' with a percentage. The 'Format' pane on the right is open, showing the 'Visual' tab. In the 'Options' section, 'Apps # Prior Cycle' is added to both the 'Upper bound' and 'Lower bound' buckets. A purple circle highlights the 'Apps # Prior Cycle' measure in the 'Data' pane on the right.

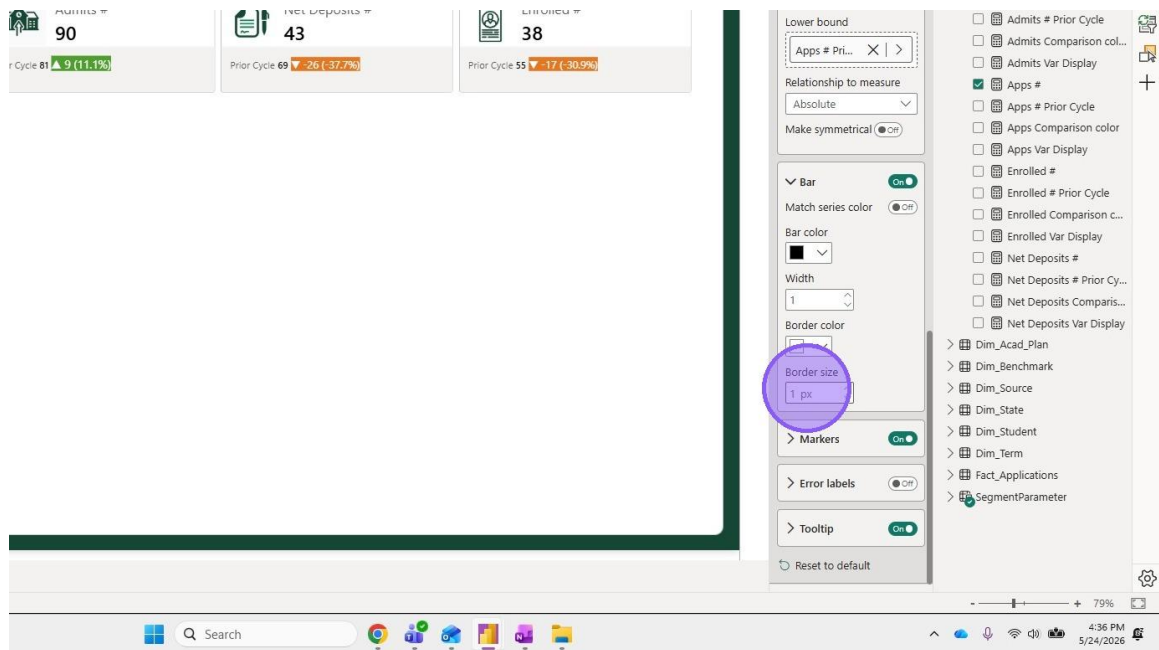
56. Under Bar, Change bar color to black

The screenshot shows the Power BI Desktop interface. The main view displays three cards: 'Admits #' with a value of 90, 'Net Deposits #' with a value of 43, and 'Enrolled #' with a value of 38. Each card shows a change from the 'Prior Cycle' with a percentage. The 'Format' pane on the right is open, showing the 'Visual' tab. In the 'Bar' section, the 'Bar color' is set to black. A purple circle highlights the 'Bar color' dropdown in the 'Format' pane.

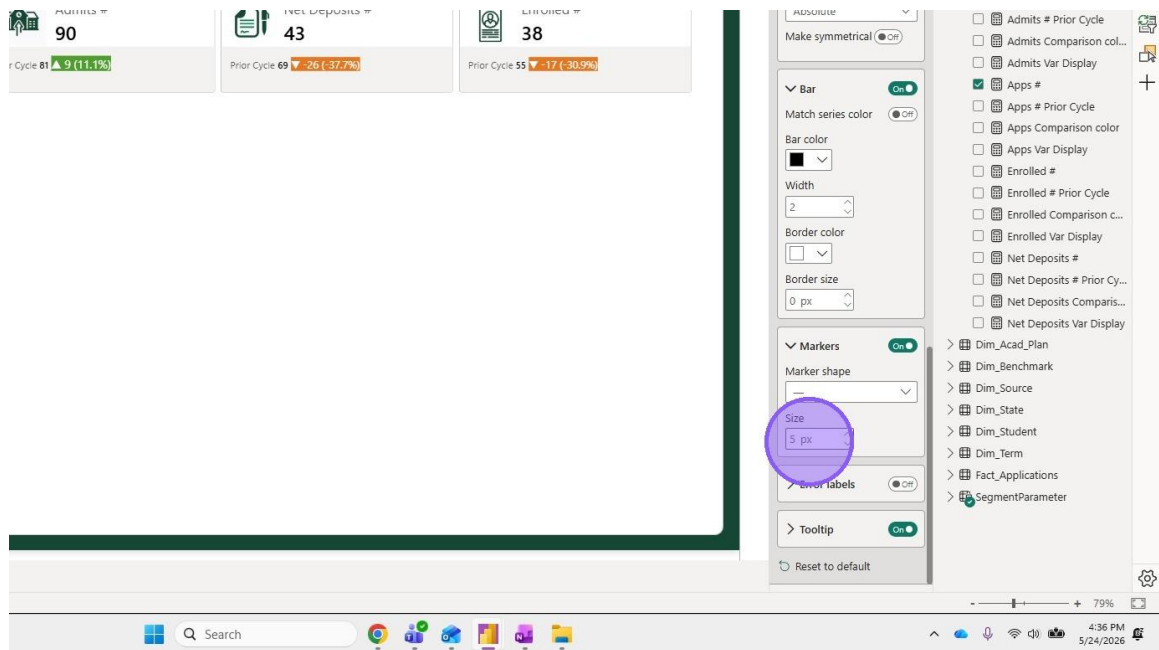
57. Change width to 2 px



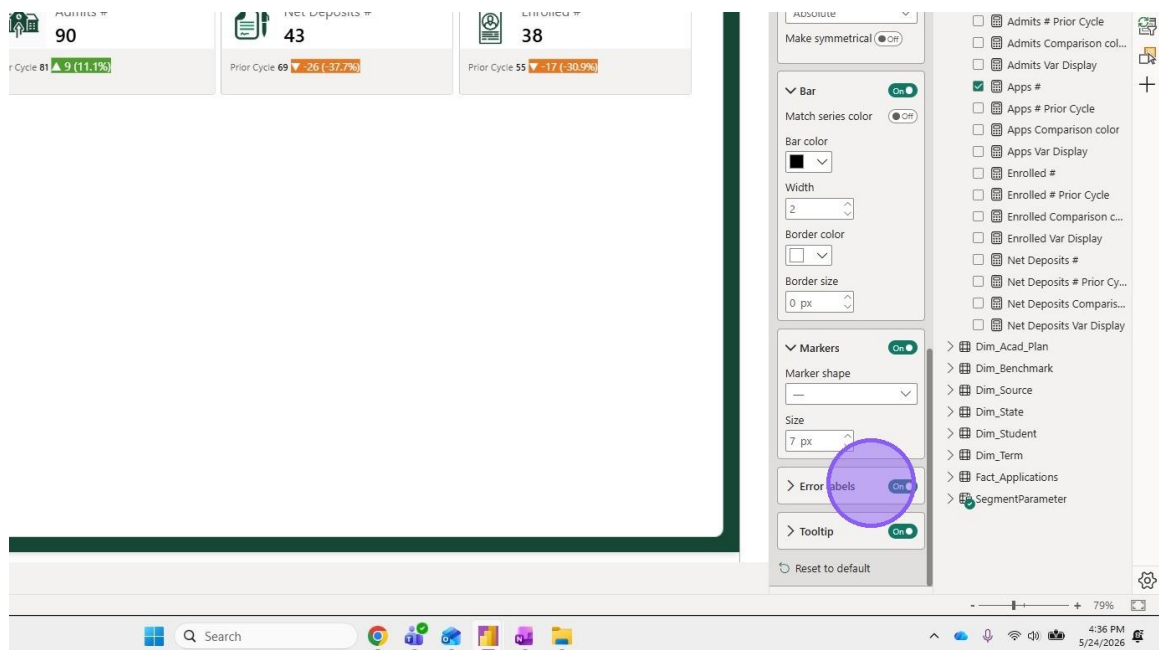
58. Change Border size to 0 px



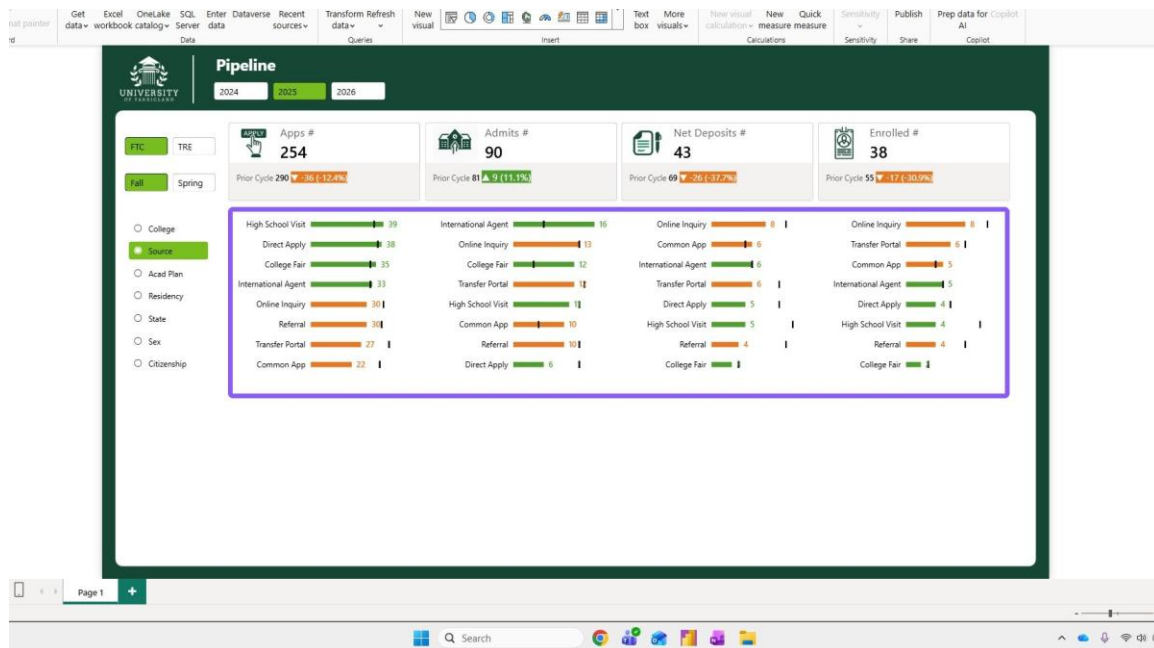
59. Click "px"



60. Optional! Turn on "Error labels" if needed

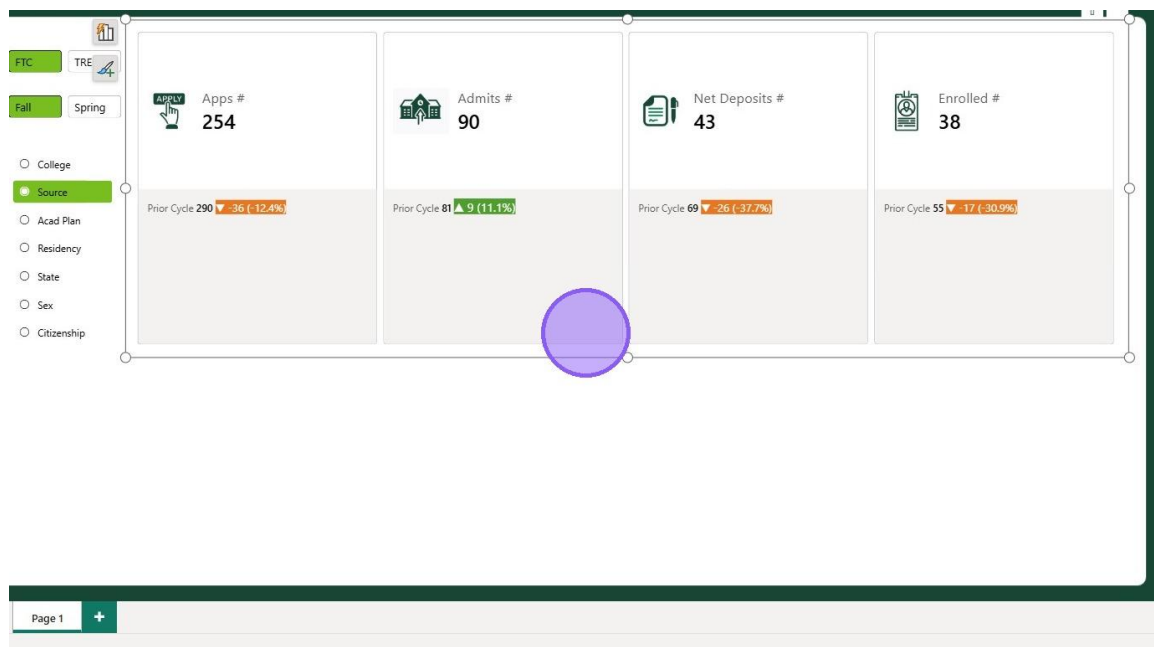


61. Repeat the steps 53 to 60 for admits #, net deposits # and enrolled #. Don't forget to change the measures for each. You should see something like below.

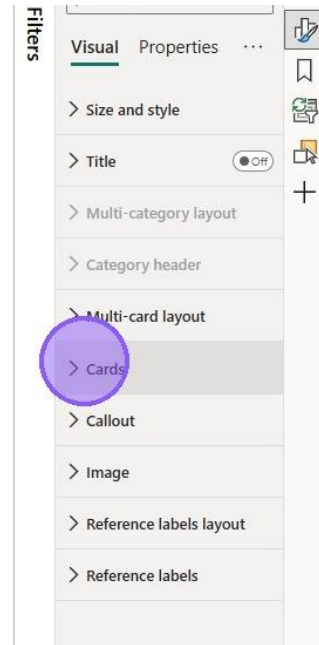
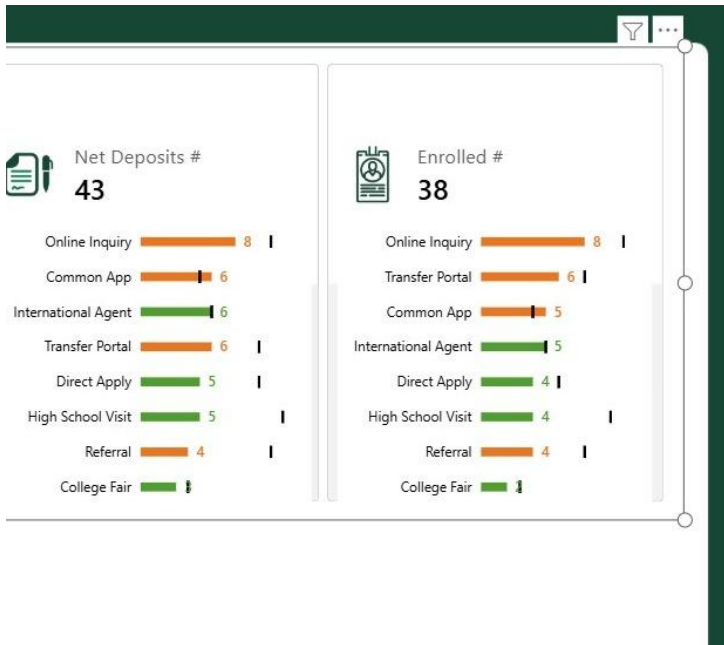


Visual Enhancements

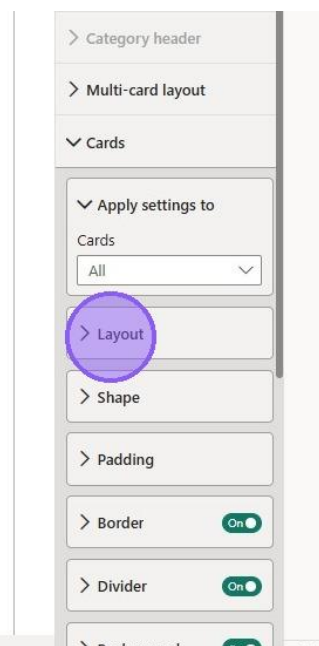
62. Drag the KPI Card to overlap the bar charts like below



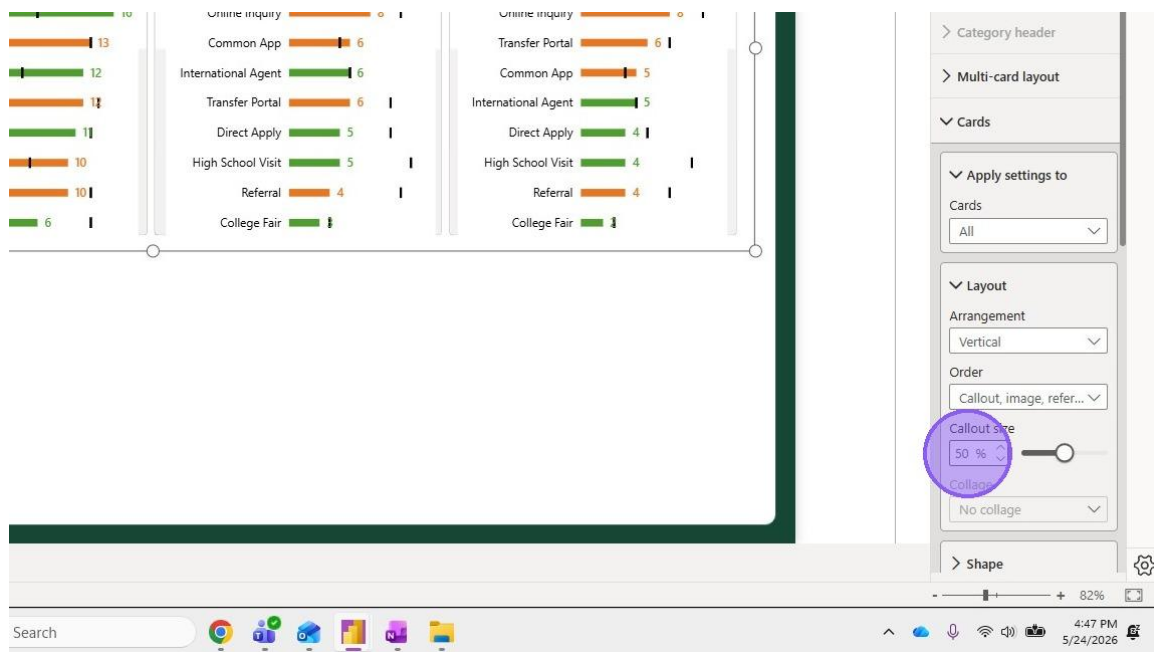
63. Click "Cards"



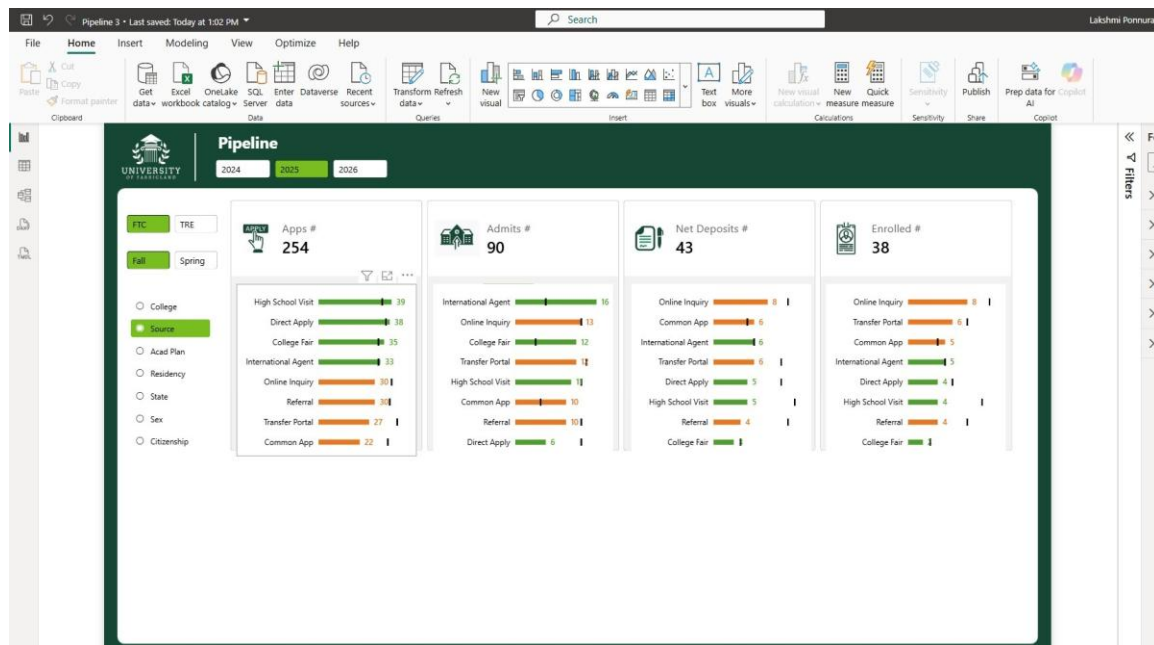
64. Click "Layout"



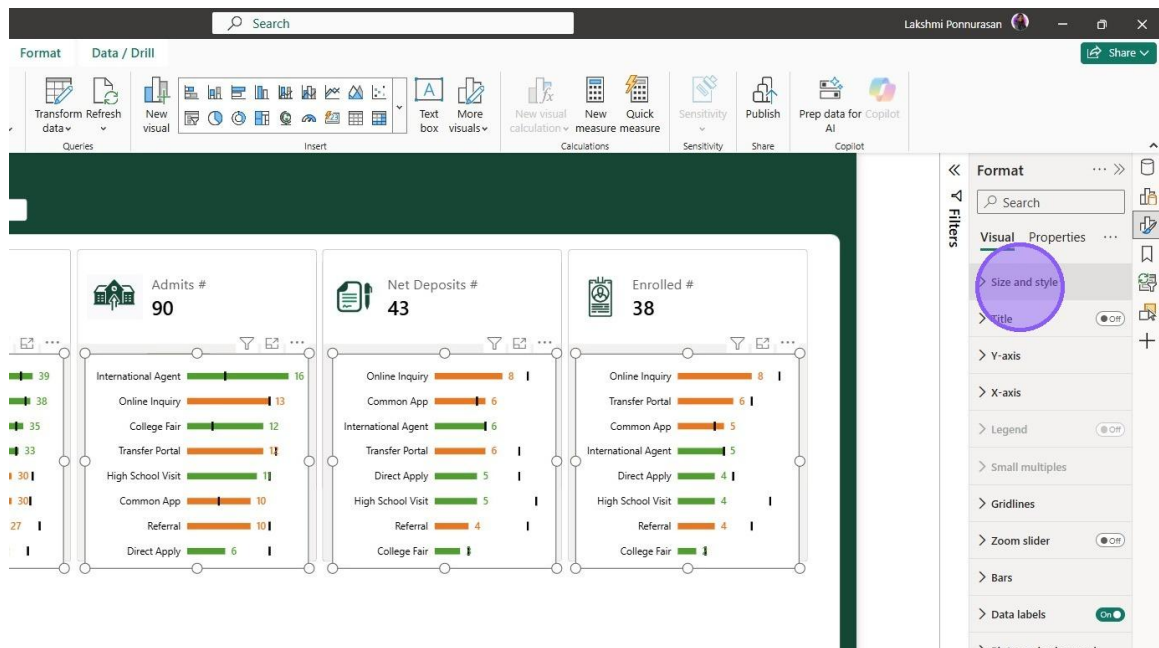
65. Change Callout size to 20 %



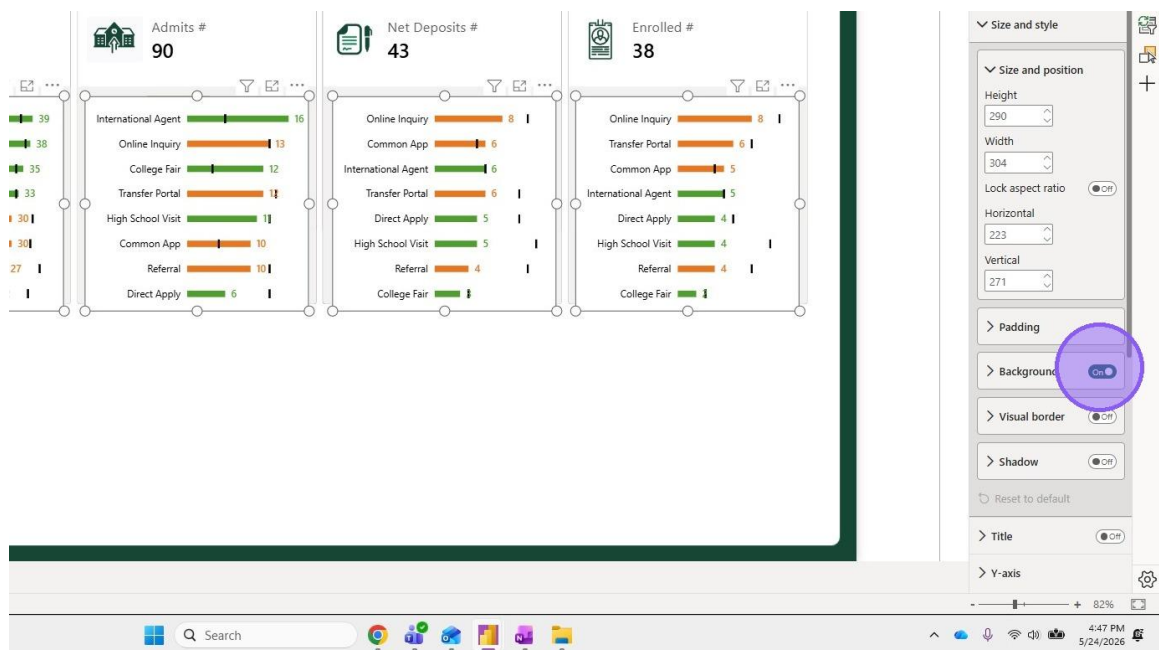
66. You should see something like this!



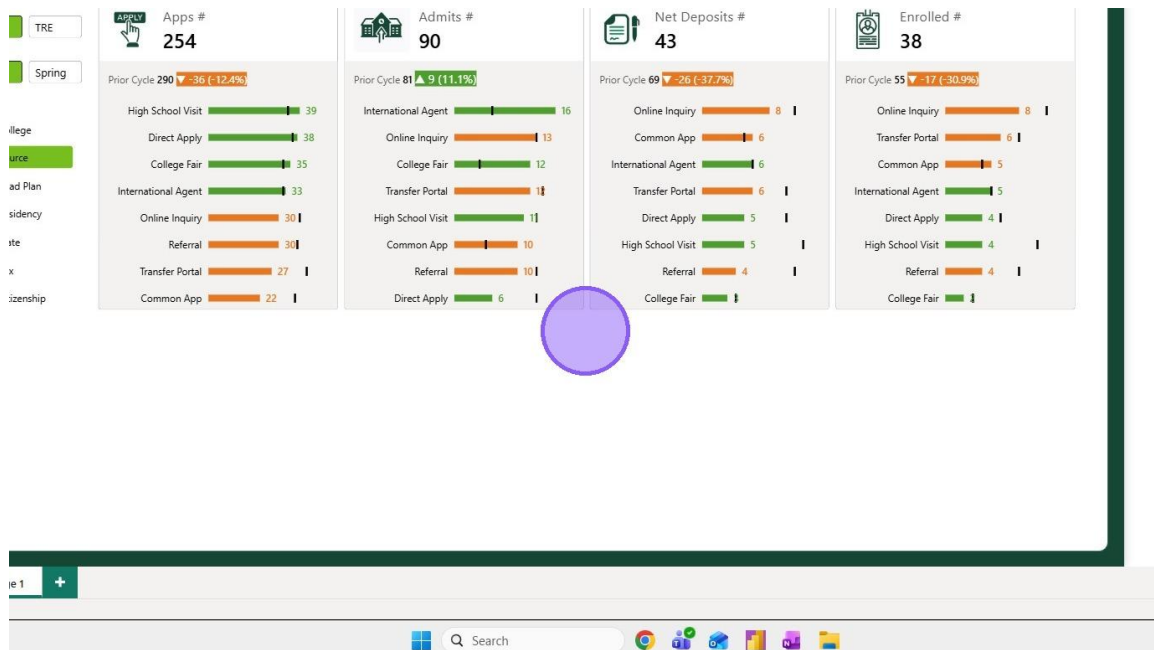
67. Click "Size and style"



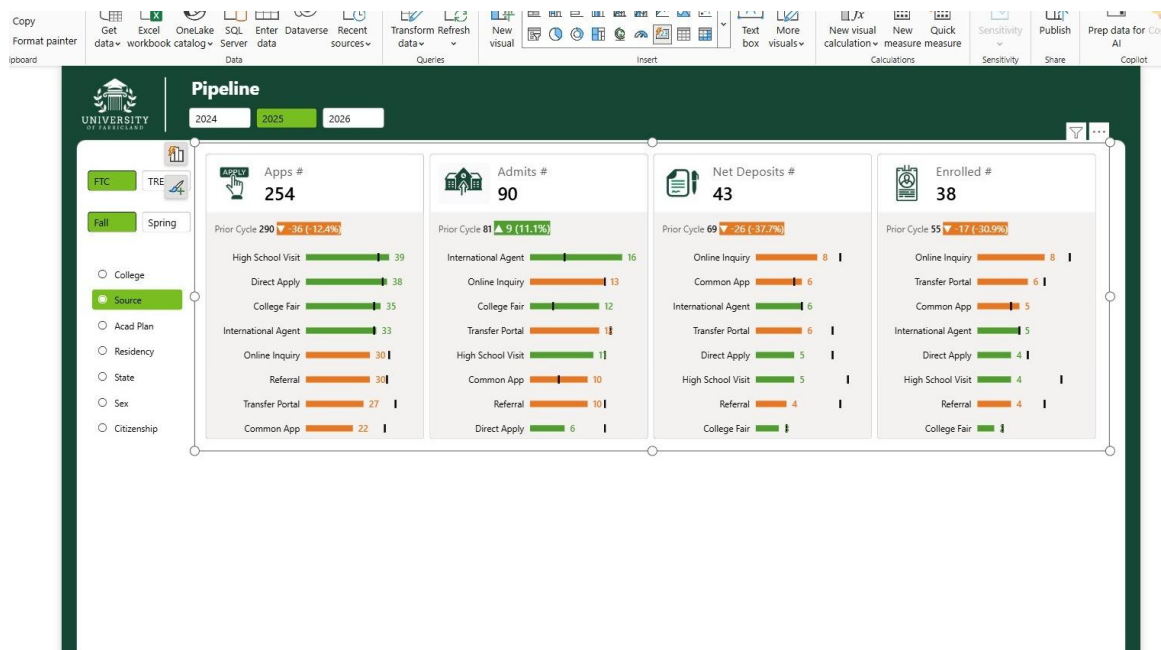
68. Click "On"



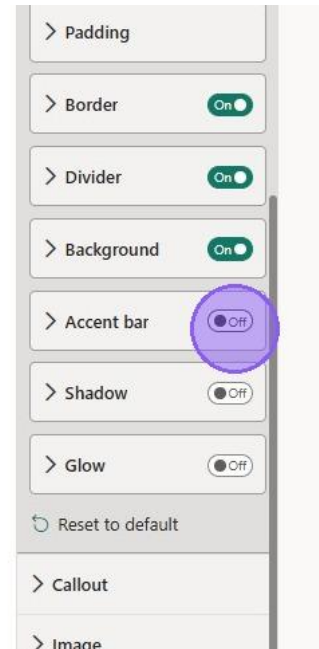
69. Click "Power BI Report"



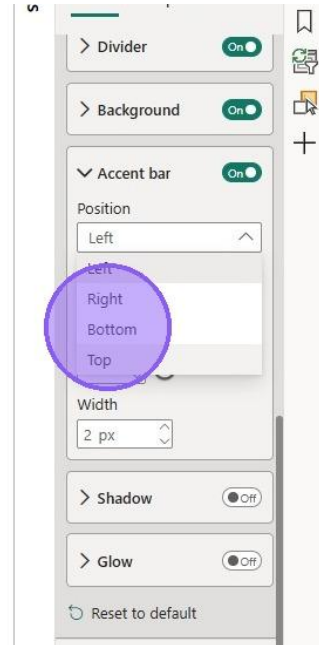
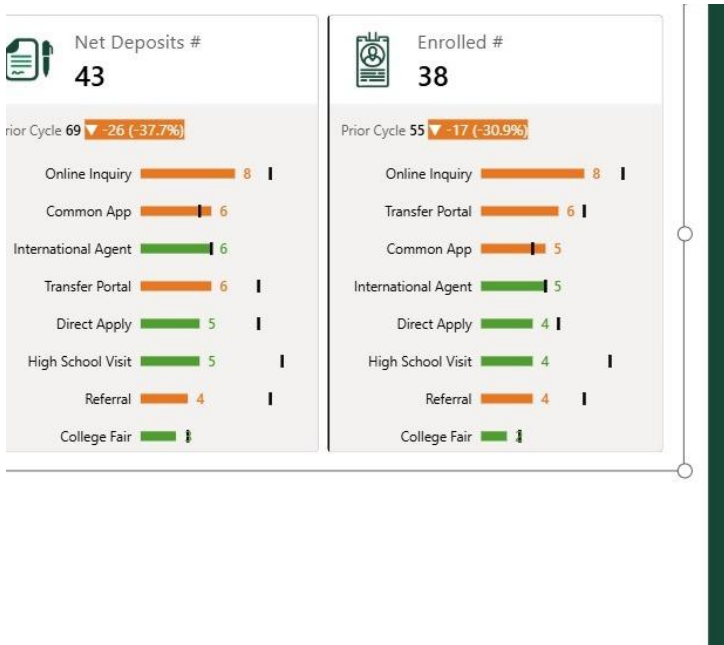
70. Select all the 4 bar charts and turn off the background to see something like below.



71. Click "Off"



72. Change Position to Top



73. Change width to 5 px



74. Now under Cards, select "Apps #"



75. Click fx to conditionally format the accent bar color

Net Deposits #
43
Prior Cycle 69 ▼ -26 (-37.7%)

Category	Value
Online Inquiry	8
Common App	6
International Agent	6
Transfer Portal	6
Direct Apply	5
High School Visit	5
Referral	4
College Fair	1

Enrolled #
38
Prior Cycle 55 ▼ -17 (-30.9%)

Category	Value
Online Inquiry	8
Transfer Portal	6
Common App	5
International Agent	5
Direct Apply	4
High School Visit	4
Referral	4
College Fair	1

Conditional formatting
Make this property change under different conditions that you define.

Visual Properties
> Divider On
> Background On
▼ Accent bar On
Position: Top
0 %
Width: 5 px
> Shadow Off
> Glow Off
Reset to default

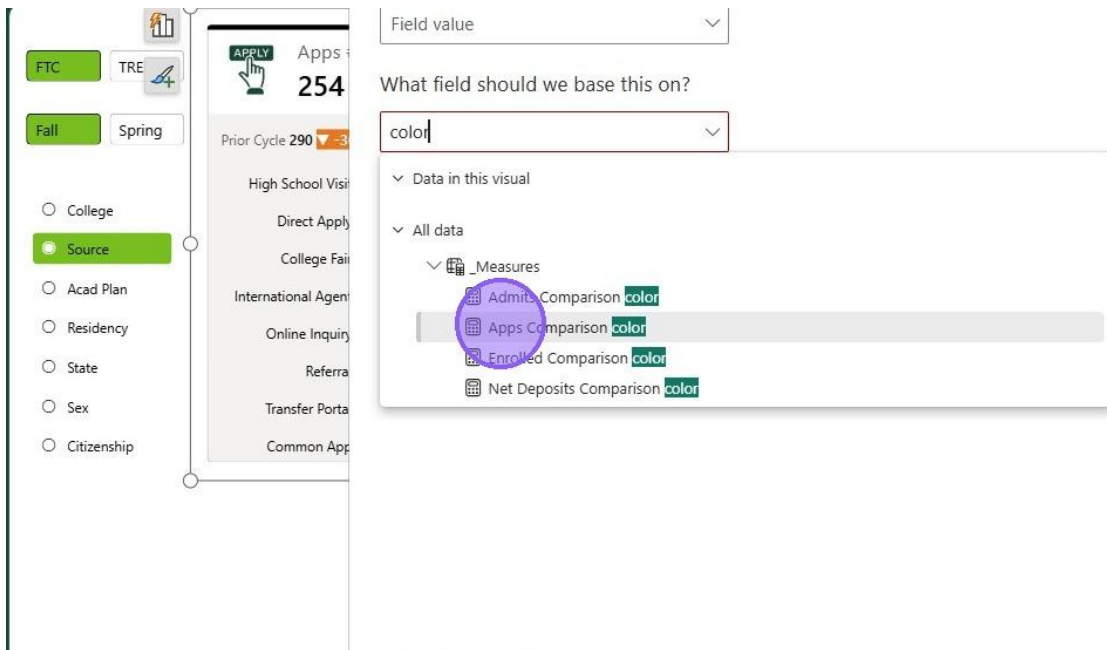
76. Click "Field value"

Pipeline
2024 2025
Applicants 254
Prior Cycle 290 ▼ -23

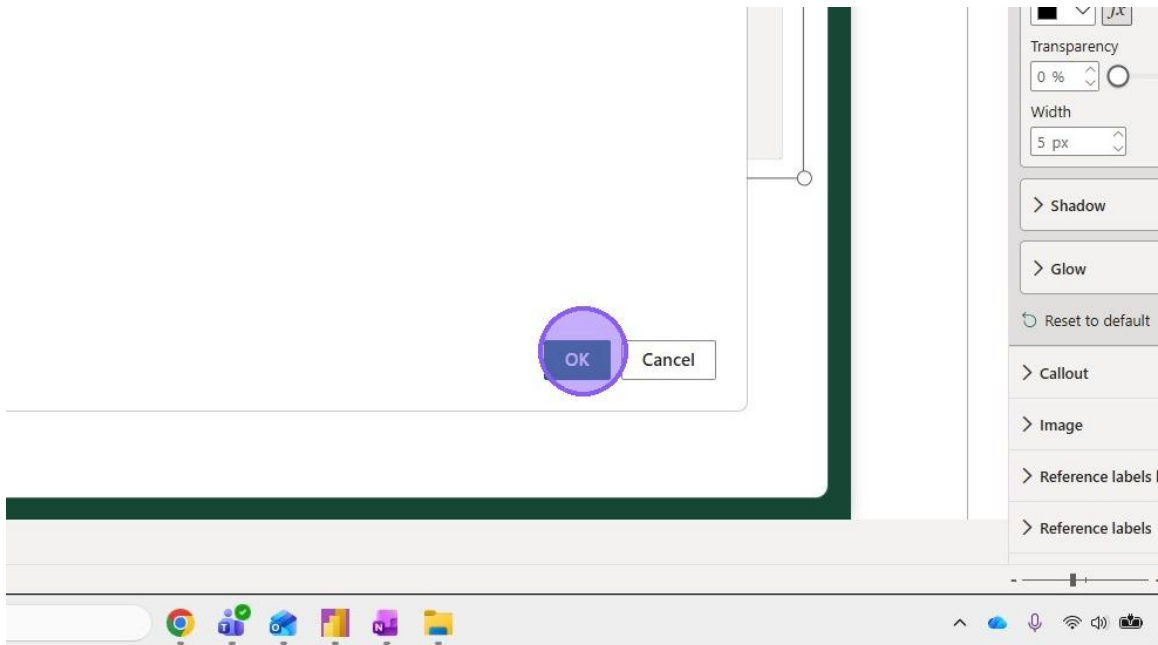
High School Visit
Direct Apply
College Fair
International Agent
Online Inquiry
Referral
Transfer Portal
Common App

Color - Cards
Format style
Gradient
Gradient
Rules
Field value
Minimum
Custom
Enter a value
Add a middle color

77. Click "Apps Comparison color"



78. Click "OK"



79. Repeat the same process for all the other cards to get something like this! Your top KPI section with segment analysis is ready!!

